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جهاز الضرائب
TAX AUTHORITY

Oman

Solution Reference Architecture

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Document History

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1.0.1	2026.05.22	Roger Young / Oman Team	Update for QR code - with implementation information for UUIDv5 Update process IDs, published on confluence



1 Introduction

1.1 Overview and Context

The following solution has the Solution view of an Access Point (AP) acting as Oman accredited Service Provider (SP) both as sending AP and receiving AP.

1.2 Architecture Scope for Oman Implementation

This document outlines the solution architecture for the implementation of Peppol in Oman, designed in alignment with Peppol, using the Oman PINT Invoice, and the national OM Tax Data Document (TDD) specification - based on the Peppol TDD (with location variations). The architecture adopts a decentralised 5/6-corner model (the SMP will be centralised by Oman Peppol Authority), enabling secure, standards-based exchange of structured invoice and tax data between economic operators, service providers, and the Oman tax authority.

The Oman Peppol ecosystem will leverage a dedicated portal - to facilitate service provider and end-user registration and Oman SMP management (through APIs).

The Oman Peppol solution will also support B2C transactions, for consumers who are not registered with Peppol.

1.3 Brief Guide to this document

This document serves as the definitive source of technical solution architecture for the Peppol eInvoicing solution in Oman. It builds upon the foundational principles and structures defined in the Peppol eDelivery framework. These references are used to avoid duplication and maintain alignment with Peppol.

The structure of this document follows a top-down progression, beginning with generic Peppol architecture elements and evolving toward the specific design and implementation of the Peppol Oman Solution Architecture.

Chapters 2 provides an overview of the architecture, highlighting the relationships and dependencies between components. These chapters also introduce the decentralised 5/6-corner model adopted in Oman, and describe the end-to-end message choreographies, including the use of the Tax Data Document (TDD) and PINT semantic structures.

Chapter 3 describes the Portal, end-user registration and SMP management.

Chapter 4 describes the B2C solution and the associated QR code.

Chapters 5 through 7 examine the major architecture roles - C2, C3, and C5 - from their respective operational perspectives:

- Chapter 5 focuses on C2 (economic operators)
- Chapter 6 on C3 (access points and intermediaries)
- Chapter 7 on C5 (tax authority endpoints and validation services).



Chapter 8 provides information on the C6 TDD consolidation and the TDD status

Chapter 9 provides a matrix of the relevant document (PINT, TDD and MLS) flows.

Chapter 10 presents the identifiers and specifications for the artefacts used within the solution, including message formats, transport profiles, and metadata structures.

Chapter 11 covers transmission protocols and routing logic.

Chapter 12 addresses security mechanisms.

Appendix A lists references used in developing this document

Appendix B defines key terms used throughout the document

Appendix C SLR/SLA timings

Appendix D QR code implementation



2 Peppol Logical Architecture View

As the generic foundation of this Solution Reference Architecture, the Conceptual and Logical Peppol Architecture follows below. This chapter presents the generic concepts used as a starting point in the subsequent chapters.

The foundational elements of the Peppol Network Conceptual Exchange Architecture is shown in Figure 1 Peppol Network Conceptual Exchange Architecture below:

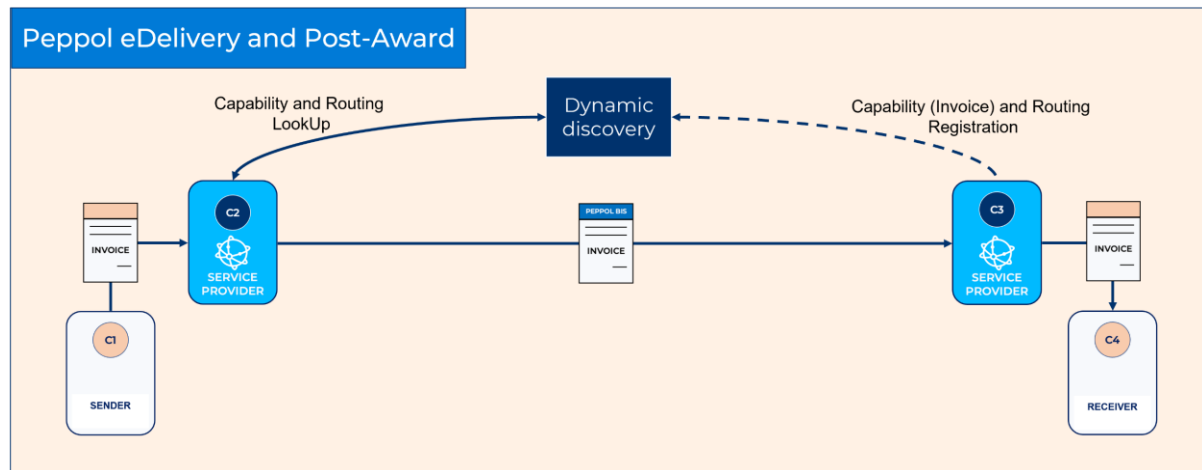


Figure 1 Peppol Network Conceptual Exchange Architecture

The exchange is divided into two phases:

1. Exchange Prerequisite: Capability and addressing registration (one-time setup per receiver)
2. Business Document Exchange: Choreography of Sending a Business Document from C1 via C2 and C3 to C4 using Dynamic Discovery.

The following sections discuss the application of this generic document exchange pattern to Business Documents (including invoices) and Message-Level Status (MLS) messages.

2.1 Business Document Exchange

In the context of a particular Business Document Exchange, the components shown in Figure 2 are used.

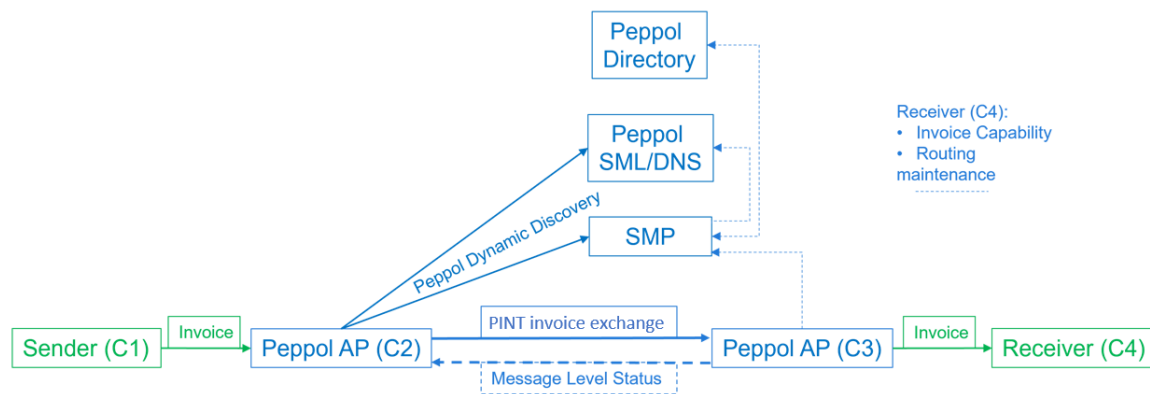


Figure 2: Peppol Logical Exchange Architecture

The components shown in blue including C2 and C3 fall into the scope of standardisation within the Peppol network; interaction between C1/C2 and C3/C4 are bilaterally agreed between the respective parties. The standardised exchange architecture isolates the choices on sending and receiving side from each other, enabling optimization independent of each other while enabling competition between access points.

2.1.1 Business Document Prerequisite

As a prerequisite for any document exchange, C3 has to register Business Document receiving capabilities of C4, but also the AP (C3) end-point and the public key of the C3 Certificate.

2.1.2 Normal Business Document flow

The document flow for Oman PINT Invoices / Credit Notes over the Peppol network is shown in Figure 3: Normal Business Document flow below.

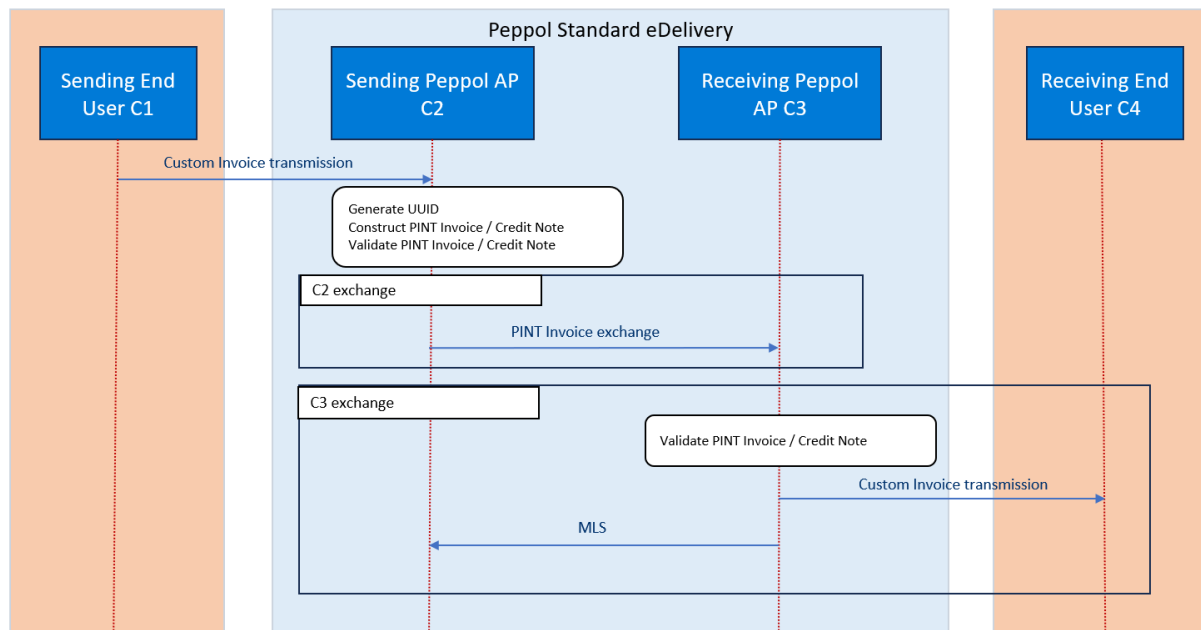


Figure 3: Normal Business Document flow

The model shows the 4-corner model, with the invoice data (in an agreed format) moving from the supplier C1 to the supplier's AP C2. C2 then prepares and validates the Oman PINT invoice and sends to the buyer's AP C3, C3 validates the invoice then sends the final invoice (in an agreed format) to the buyer (C4).

2.2 Overview of CTC Logical Architecture

The Oman solution will also include sending tax reports to the OTA. The Logical Architecture of this 5-corner model can be divided into three different exchanges:

1. Invoice exchange: C1 → **C2** → **C3** → C4 transmitting an Invoice
2. Tax Data exchange: (C1 →) **C2** → **C5** → C6 transmitting a TDD
3. Tax Data exchange: (C4 ←) **C3** → **C5** → C6 transmitting a TDD

Where each exchange above can be seen as normal Peppol eDelivery exchange of a Business Document in the middle of the standardised 4-corner model. This means that general requirements, trust model, dynamic discovery, transmission from the Peppol 4-corner model are reused.

The IT Architecture is shown in Figure 4: Full model with exchange from C1 to C6 below:

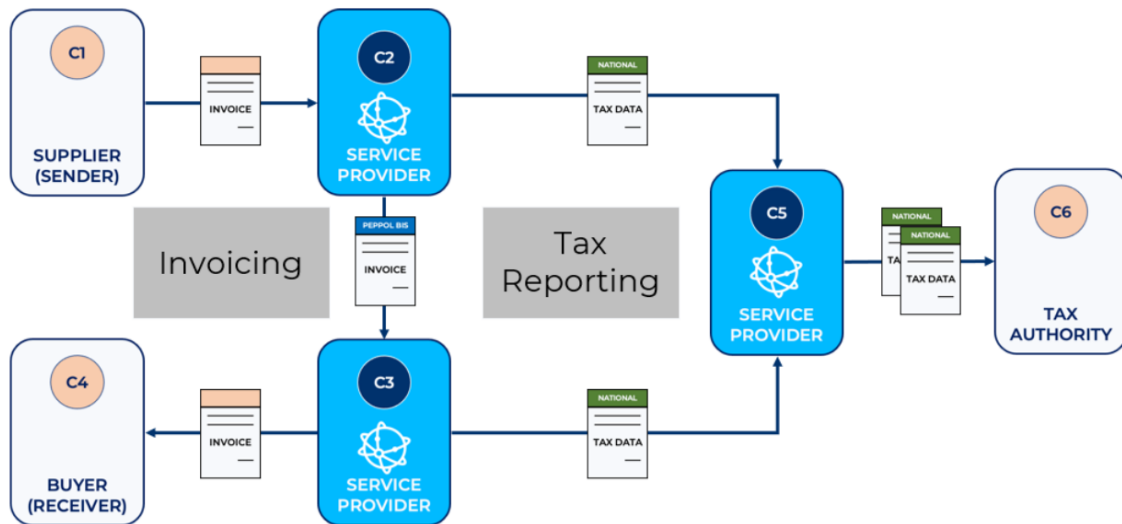


Figure 4: Full model with exchange from C1 to C6

Note: Sequencing of the C2->C3 invoice and C2->C5 TDD submission are in parallel.

2.3 Message-Level Status (MLS) Exchange

In the addition to direct synchronous feedback on the protocol layer, the Peppol network standardised message handling requires communication on its result between the participating access points (AP). This is distinct from any business-level follow-up exchanged between the End Users as the business process continues.

The main motivation is to share information within the standardised exchanges to avoid manual research and operational interaction in case of compliance deviations.

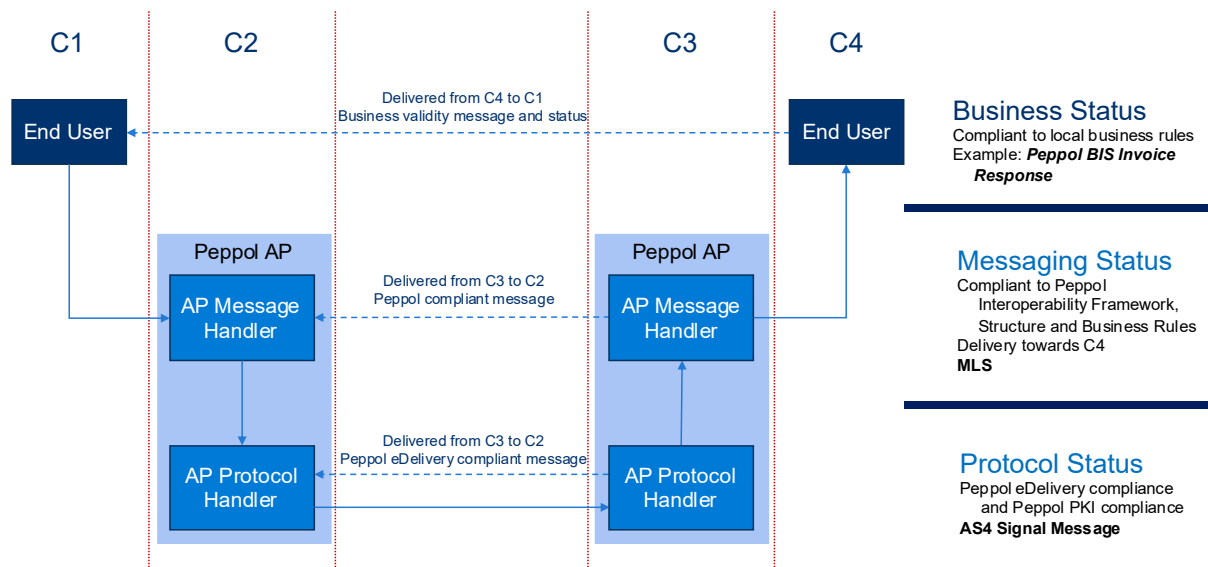


Figure 5: MLS placement in Response/Status hierarchy



As seen in Figure 5: MLS placement in Response/Status hierarchy above, the invoice MLS is a status between C3 and C2 covering the Receiver (C3) status of the compliance in relation to Peppol Business Document Specifications and onwards delivery towards C4.

While the Business Document flows from C1 to C4, the invoice MLS is from C3 to C2. It is transmitted using the same Dynamic Discovery and transmission method as a Business Document (note: the Corners from the Original Business Document exchange are being used i.e. here C3 as Sender and C2 as Receiver of MLS). Shown in Figure 6:

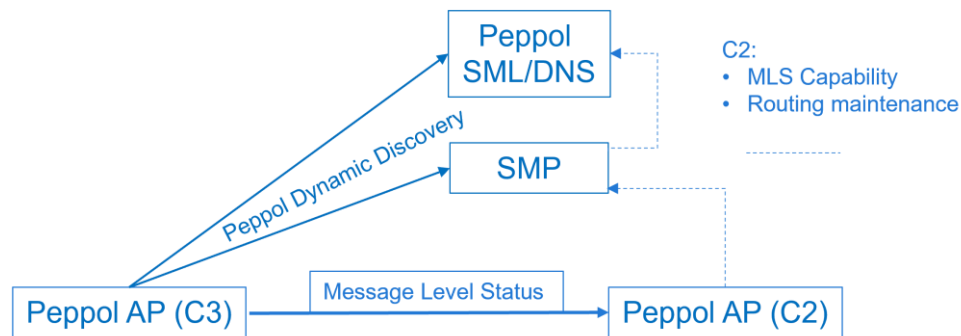


Figure 6: Peppol MLS Exchange Architecture

The TDD MLS is also used by C5 to report status of the tax report, sent from C5 to the sending AP (C2 / C3). Shown in Figure 7: TDD MLS eDelivery.

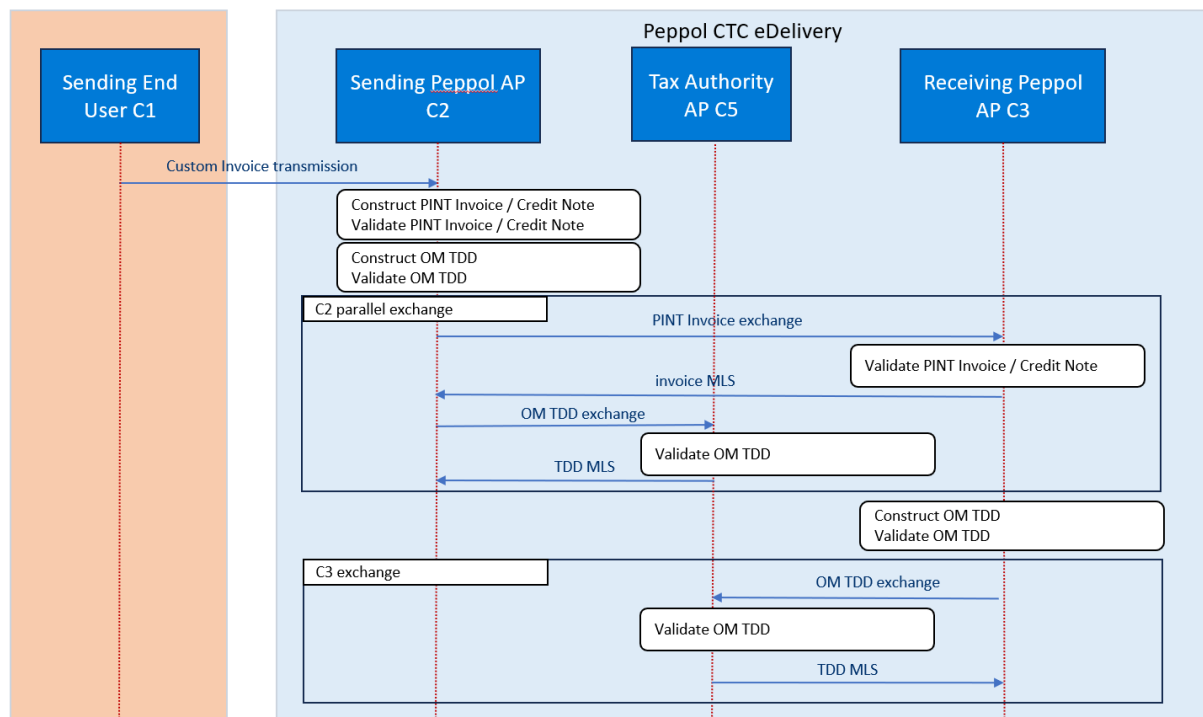


Figure 7: TDD MLS eDelivery



2.3.1 MLS Prerequisite

As prerequisites for any MLS exchange:

- C2 / C3 has to register its MLS receiving capabilities and receiving endpoint, along with its certificate.
- C2 / C3 is able to add MLS **ALWAYS_SEND** in the SBDH of the transmission (see Peppol MLS Specification) when sending business documents.

The components C2 and C3 fall into the scope of standardisation within the Peppol network. In the case of the MLS exchanges, only standard components are involved:

The invoice MLS is sent from C3 to C2 to confirm the status of the invoice submission from C2 to C3.

The TDD MLS is sent from C5 to C2 / C3 to confirm the status of the TDD submission from C5 to the sending AP.

The response codes for MLS are shown in the below table:

Code	Name/Description (Peppol Docs)	Meaning for 5 corner model (see note)
AP	Message delivered towards C4 with confirmation	Delivery succeeded for C3/4 and C5/6 and confirmation was received.
AB	Message delivered towards C4 without confirmation	Delivery succeeded for C3/4 and C5/6, but no confirmation was received.
RE	Message rejected or delivery towards C4 failed	Delivery failed, or message was rejected at C3 or C5.

Note: The 5/6-corner model is two 4-corner models, and the MLS codes support the 4 corner model so they can also be used by C3 or C5 (the receiving Access Point will know who sent the MLS):

1. Invoice exchange: C1 → **C2** → **C3** → C4
2. Tax Data exchange: (C1 →) **C2** → **C5** → C6
3. MLS (TDD) exchange: (C4 <-) **C3** <- **C5** | C6

2.4 Peppol Network Specifications

The Peppol Post-Award Architecture is defined by these generic Peppol Specifications:

Peppol eDelivery Network [eDN]

- Peppol Discovery
- DNS/SML [SML]
- SMP [SMP]



- Peppol Transmission (AS4)
- Peppol MLS [MLS]

Peppol Code Lists [CL]

Peppol Code Lists [CL] provide the authoritative references for identifiers and values used within the Peppol eDelivery Network, ensuring consistency and compliance across Access Point implementations [APL]

Peppol Policies and Specifications relevant for this context:

- [eDN] Peppol eDelivery Network documentation: <https://docs.peppol.eu/edelivery/>
 - Peppol Discovery
 - SML
 - SMP
 - Peppol Transmission
 - AS4
 - MLS
 - Peppol Policy for use of Peppol Network (PPfuoPN)
 - Peppol Policy for Use of Identifiers (PPfuol)
- [CL] Peppol eDelivery Code List: <https://docs.peppol.eu/edelivery/codelists/>
- [APL] List of sample AP products: <https://peppol.org/tools-support/links-to-software/>
- [PoA] Peppol Post Award Documentation: <https://peppol.org/documentation/technical-documentation/post-award-documentation/>
- [PINT] and [TDD] (Draft) Oman Specifications for PINT: <https://test-docs.peppol.eu/pint/pint-om/2026-Q2-MR/>



3 Portal

Oman's eInvoicing framework includes a portal responsible for managing service provider accreditation related processes, and processes for managing the C1/C4 and service provider relationships. Oman will also maintain its own centralised SMP [SMP], which will be used for all domestic participant records.

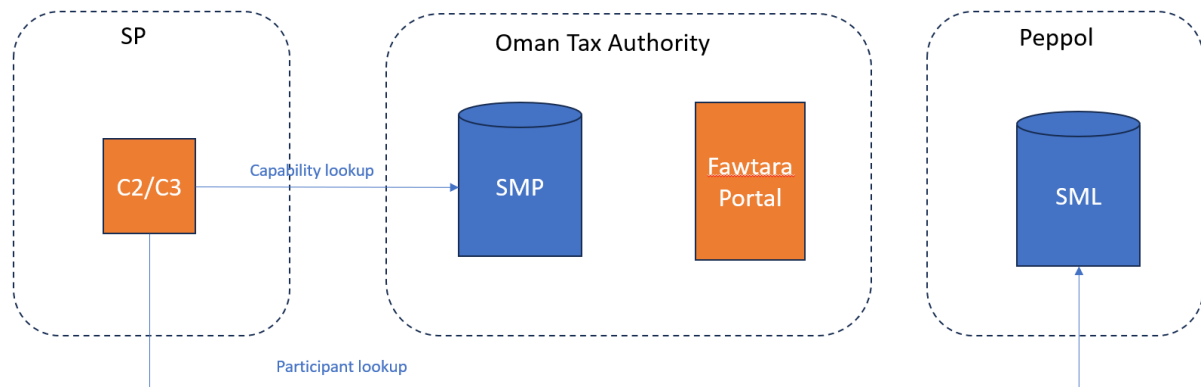


Figure 8: Oman SMP / SML participant lookup

3.1 Service provider and TP setup

Service providers must access the Fawtara Portal via Single Sign-On (SSO) using their Tax Authority portal credentials, they can then submit an application to express interest to become an access point.

Upon receiving full accreditation, the service provider is listed on the Fawtara Portal as an accredited service provider / access point for Oman. When the C1/C4 accesses the Fawtara Portal, also via Single Sign-On (SSO) using their Tax Authority portal credentials, they can select an accredited service provider to act as their Access Point for the Peppol eDelivery network. This relationship is initiated by the TP sending a connection request to the AP via the portal.

3.2 Relationship management

The Oman Peppol Authority has developed mechanisms to manage taxpayer-service provider relationships on the Fawtara Portal and Oman SMP, which include the portal:

- Providing a trust mechanism linking taxpayers to accredited service providers.
- Allowing taxpayers to connect, disconnect from and reconnect with service providers.
- Enabling service providers to effectively manage their connections with taxpayers.
- Facilitating participant management with the Oman SMP system through the provision of SMP management APIs.

Note: Service providers (C2/C3) must be accredited by the Oman Peppol Authority before they can receive portal link requests from taxpayers (C1/C4).



Steps for Tax Payer / Portal / SMP / SML registration:

1. Step 1

- a. Service Provider (**SP**) accesses Portal to register interest in accreditation
- b. OTA and **SP** complete the accreditation processes
- c. SP is listed in Portal as an accredited access point

2. Step 2

- a. The Tax payer (TP) C1/C4 accesses the Fawtara Portal and selects a **SP**
- b. The **SP** receives a request from the **Portal** to accept the TP request
- c. The **SP** accesses the portal to accept the request
- d. The **SP** then contacts the TP for onboarding (outside Portal) to agree:
 - i. receiving capabilities (PINT INV / CN, SB INV / CN etc.) *
 - ii. TP's Tax authority (see section 10.3.2)
 - iii. C1-C2 Bi-lateral data submission format (outside Peppol)
 - iv. C3-C4 Bi-lateral data submission format (outside Peppol)

3. Step 3

- a. The SP registers the TP Participant ID (PID) and Service Group (SG) on the SMP
 - i. This step is validated by the portal (see Figure 9: Oman Portal / SMP architectural overview)
- b. The SP updates the SMP SG with TP receiving capabilities (PINT INV / CN, SB INV / CN etc.)
 - i. This step is validated by the portal (see Figure 9: Oman Portal / SMP architectural overview)
- c. **SMP** tells the SML to create the PID lookup record (see box below Figure 9: Oman Portal / SMP architectural overview)
(SML verifies uniqueness of the PID)

* If the SP and the TP do not reach an agreement before the SP accepts the request then the TP can withdraw the request. Once the SP accepts the request then a disconnection request need to be initiated in the portal by the TP.

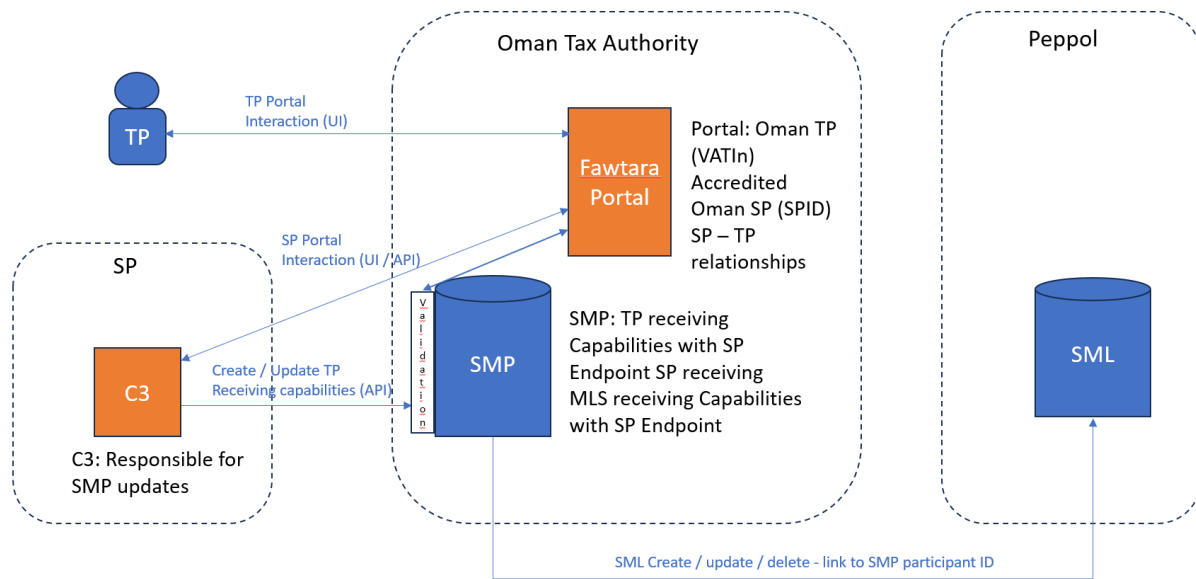


Figure 9: Oman Portal / SMP architectural overview

SML participant lookup record

After updating the Taxpayer's metadata (PID and SG) in the OM SMP, the OM SMP then sends a SOAP request to the SML's management interface:

- createBusinessIdentifier
- deleteBusinessIdentifier
- updateBusinessIdentifier

This request contains only:

- The Participant ID
- The Oman SMP's domain name (that hosts the metadata)

The SML authenticates the request using the SP's Authorisation Certificate.

The SML creates or updates the DNS entry:

- CNAME record pointing the Participant ID to the SMP's hostname.

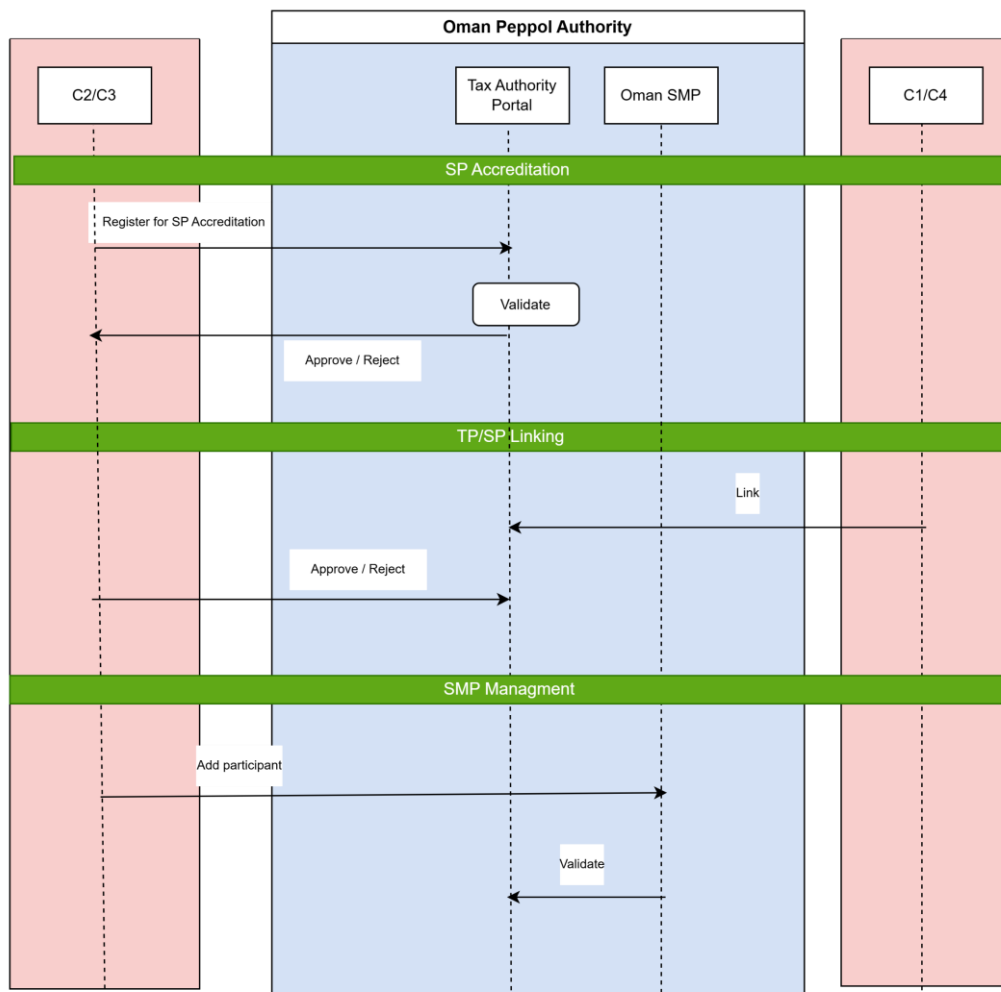


Figure 10: Oman Portal / SMP sequence diagram



3.3 Testing environment

The below figure shows the configuration of the OTA / OpenPeppol testing environment.

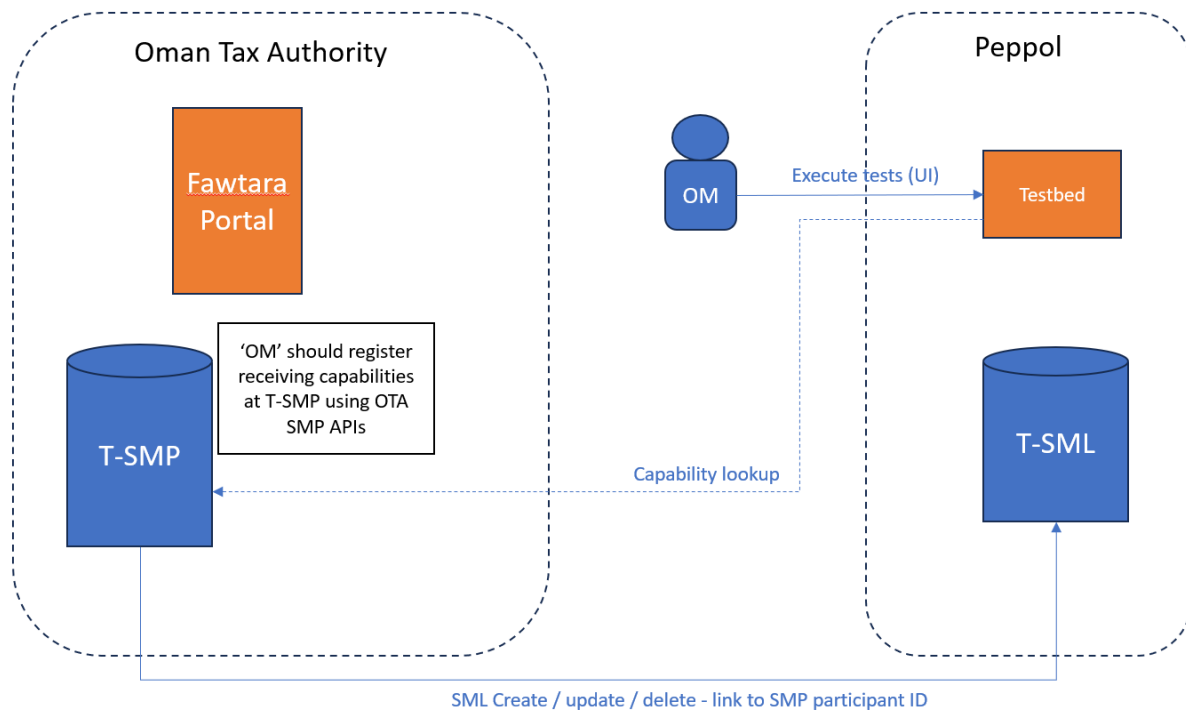


Figure 11: Oman Testing environment

3.4 Operational

The Accredited Service Provider must add the C1/C4 to the central SMP within 3 working days of establishment of a relationship on the Fawtara Portal.

When relationship on the Fawtara Portal ceases to exist/is terminated, the Accredited Service Provider has 1 working day to deregister the C1/C4 from the central SMP.

When C1/C4 decides to switch from one Accredited Service Provider to another, the existing Accredited Service Provider must deregister C1/C4 from the central SMP within 1 working day and the new Accredited Service Provider must add C1/C4 to the central SMP within 3 working days.



4 B2C and QR code

4.1 Oman B2C invoicing

The Oman eInvoicing solution includes a B2C component, this will usually be buyer receiving a human readable invoice, sent outside of Peppol, plus the submission of a TDD sent by the seller's AP, with an embedded Simplified OM PINT Invoice (see section 10.2.1).

To comply with OM PINT mandatory fields fixed values will be used for the buyer, who will not be registered (or identified as registered) in Peppol. Notes:

1. Typically, B2C sales are low value transactions.
2. Some of these transactions could be B2B registered Peppol buyers, but will still be treated as a B2C transaction. For example, a company purchasing stationary at a store.
3. For certain sales - usually involving higher valued items, more information will be collected by the seller (even for B2C), including details of the buyer. For example, a mobile phone purchase where warranty may be applicable. In this case a different invoice type would apply.
4. Profit Margin – OM PINT (not simplified) invoice (see section 10.2.1)

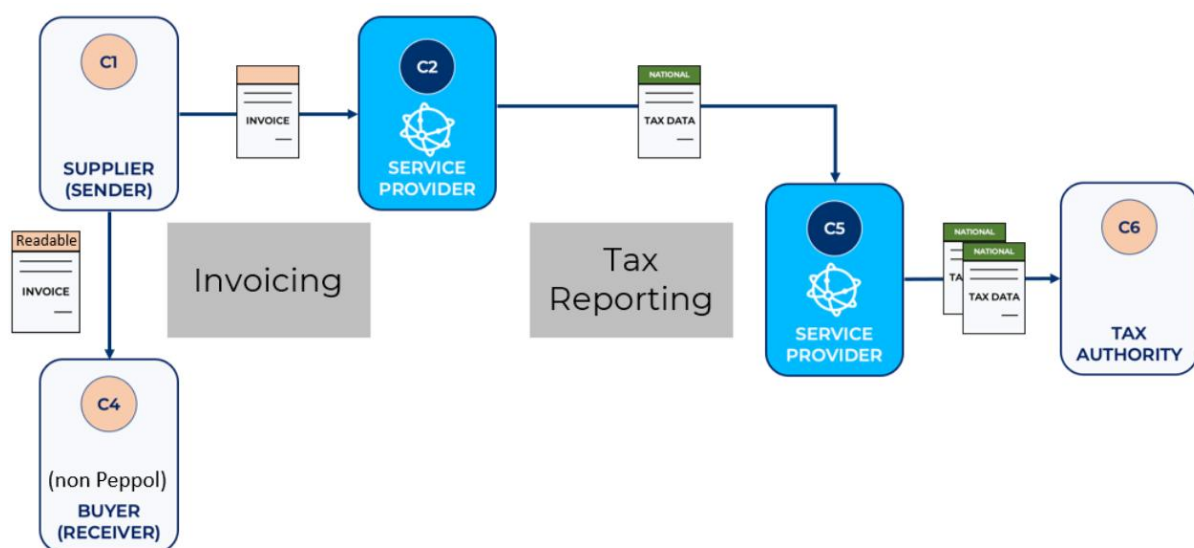


Figure 12: Oman B2C transaction diagram

4.2 QR code

In B2C transactions the seller will provide the consumer with a readable invoice (printed or via email), this invoice will include a QR code.

See invoice example:



Figure 13: Oman B2C readable invoice

The QR code on a B2C invoice (full or simplified) must contain the following:

Field	Value
QR Version	1.0 [value is fixed but subject to be changed by OTA, new version may suggest new structure]
Invoice Type	01 [Fixed value]
IBT-027 (seller name)	Name of the Seller (C1)
IBT-031 (VATIN of the seller)	VATIN of the Seller (C1)
IBT-002 (invoice date)	Invoice issuance date Date format YYYY-MM-DD
IBT-168 (time stamp)	Invoice issuance time Time format hh:mm:ss
IBT-112 (invoice total amount – incl. VAT)	The invoice total amount with VAT
IBT-110 (VAT total)	VAT total of the invoice
BTOM-004: (Seller UUID)	Seller UUID generated by C1 based on all the above-mentioned fields using the namespace defined in section 10.2.3



The QR code on a Profit Margin invoice must contain the following:

Field	Value
QR Version	1.0 value is fixed but subject to be changed by OTA, new version may suggest new structure]
Invoice Type	02 [Fixed value]
IBT-027 (seller name)	Name of the Seller (C1)
IBT-031 (VATIN of the seller)	VATIN of the Seller (C1)
IBT-002 (invoice date)	Invoice issuance date Date format YYYY-MM-DD
IBT-168 (time stamp)	Invoice issuance time Time format hh:mm:ss
BTOM-030 (Total amount due)	Total amount of the profit margin invoice including VAT
BTOM-004 (Seller UUID)	Seller UUID generated by C1 based on all the above mentioned fields using the namespace defined in section 10.2.3

See Appendix D for examples on how to generate the Seller UUID.

4.3 PINT for B2C

The B2C transactions will not use a different invoice, it will still be a PINT Invoice. The PINT has mandatory fields for the buyer, which will not be available to C1, so C2 will populate generic values for these fields so that PINT validation is successful.

B2C relevant fields:

OM PINT ID	Business term	Description	Value
IBT-044	Buyer name	The full name of the Buyer.	'General customer'
IBT-055	Buyer country code	A code that identifies the country.	'OM'
IBT-001	Invoice number	A unique identification of the Invoice.	Generated by Seller

BTOM-004	Seller UUID	Code generated by the seller (C1) and passed to C2 as part of the invoice data, for inclusion on the Invoice / TDD (new field – see 10.2.3)	Generated by seller
IBT-049	Buyer identifier	An identifier of the Buyer. Dummy participant ID (see 10.3.1) Note: For profit margin invoices the IBT-046 will be the identifier and the dummy PID will still be used	0248: 997770000099

The following business rules will need creating for B2C invoices.

Rule name	Rule description
IBR-173-OM	If Buyer identifier (IBT-049) is '997770000099', Seller UUID (BTOM-004) MUST be present.

The following Figure 14: B2C sequence diagram shows the B2C purchase through to TDD submission to the OTA.

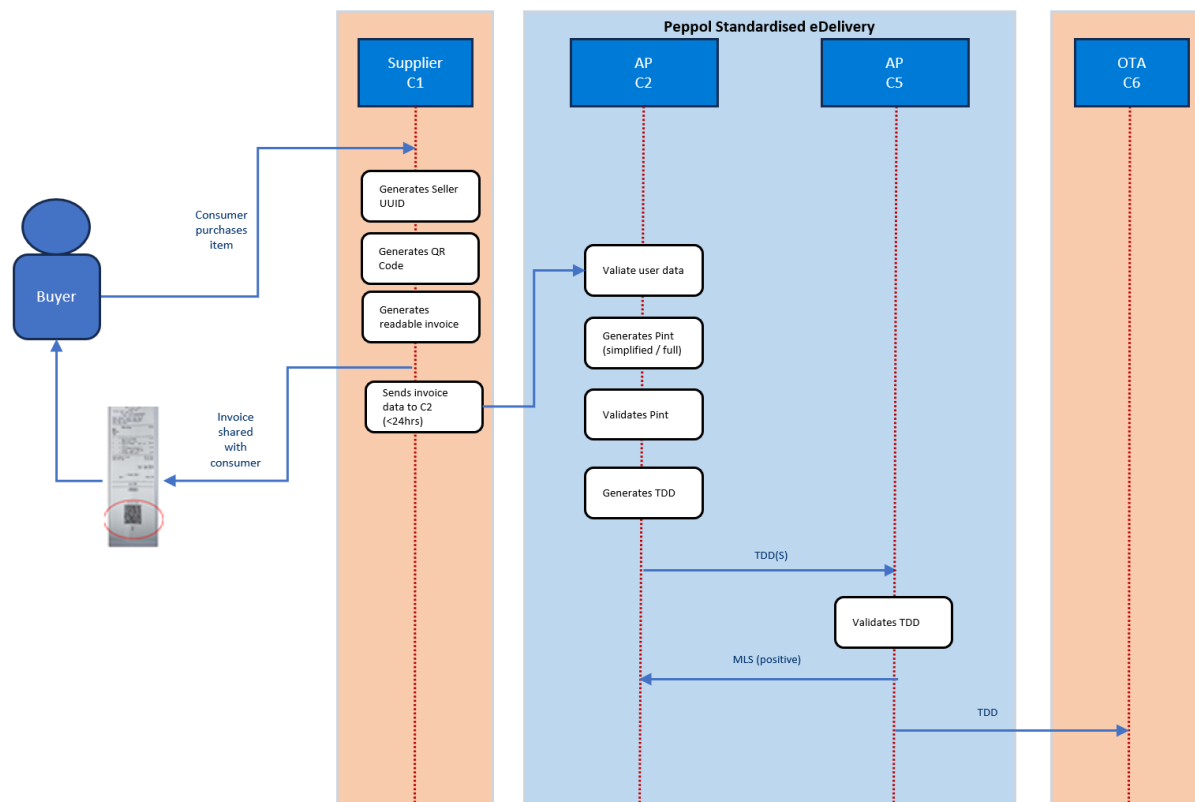


Figure 14: B2C sequence diagram

Note: the supplier should send the invoice to their AP within 24 hours.



5 C2 Capabilities

5.1 C2 Prerequisites

5.1.1 Registration in the Dynamic Discovery

C2 **MUST** register receiving capabilities in the SMP for its own appropriate endpoint by its:

- Service Provider Identification (see section 10.3.3)

With the following Document Type:

- Message Level Status (MLS) [MLS] (see 2.3)

5.1.2 Functional

C2 **MUST** be able to generate UUIDs for Invoice and TDD instances and Transmission Instances.

C2 **MUST** be able to create Invoice and Credit Note Business Documents based on Oman PINT Billing and exchange with C3.

C2 **MUST** be able to prepare an Invoice and Credit Note by enriching an Oman PINT invoice provided by C1 - by enriching the C1 invoice with UUID etc.

C2 **MAY** be able to create Self-billed Invoice and Self-billed Credit Note Business Documents based on Oman PINT Billing and exchange with C3.

C2 **MUST** be able to create an OM TDD from Invoice Business Documents and exchange with C5.

C2 **MUST** be able to prepare a TDD by enriching an Oman TDD provided by C1 - by enriching the C1 TDD with the Invoice UUID and the C1's Tax Authority Participant ID (see section 10.3.2).

C2 **MUST** be able to handle incoming MLS for invoice and TDD.

C2 **MUST** be able to confirm any C1 provided TDDs are a correct representation of the matching Invoice.



5.2 C2 flow capabilities

C2 follows the flow depicted in the below sequence diagram:

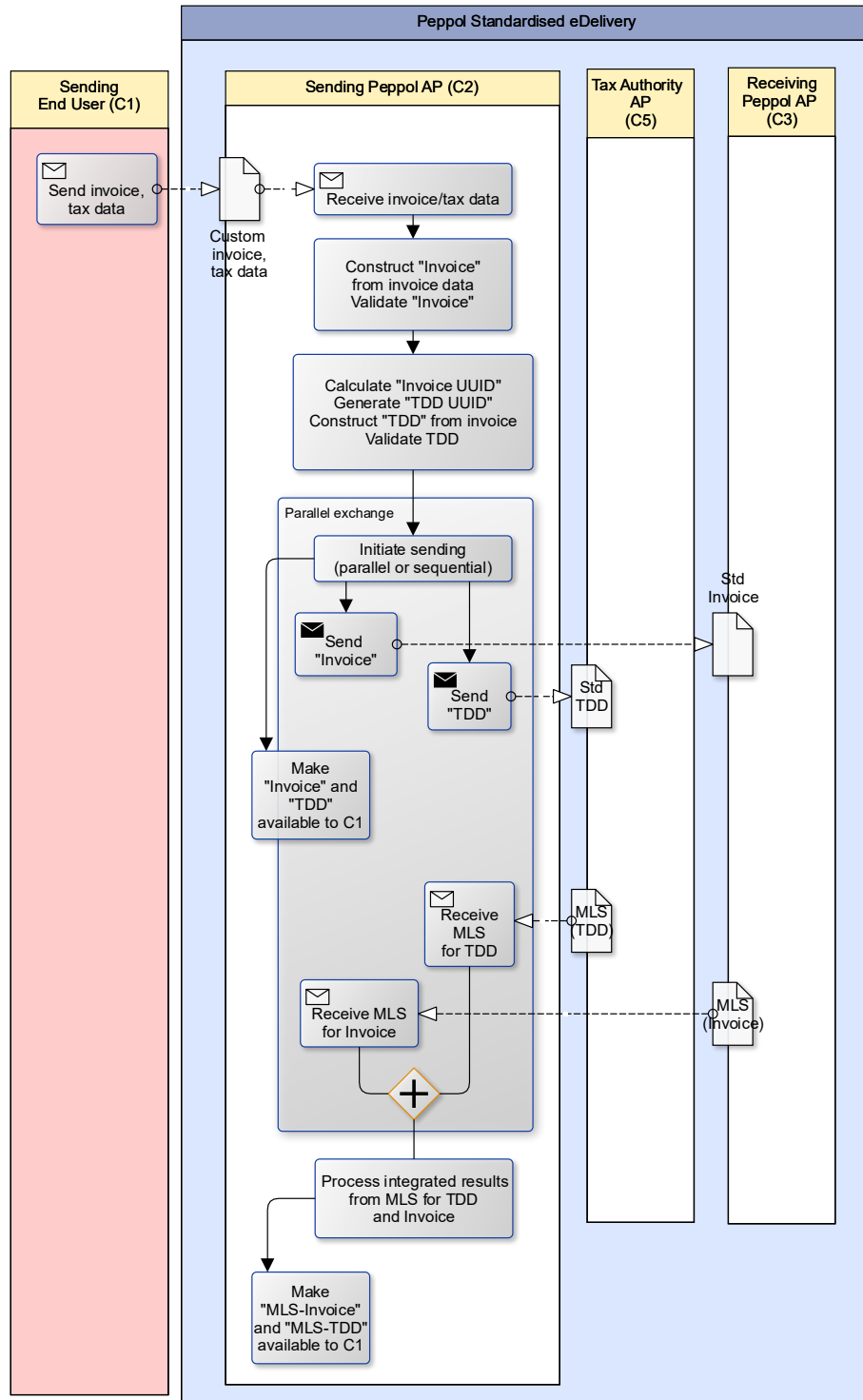


Figure 15: C2 CTC transmissions



1. C2 receives the Invoice in an internal (C1-C2) agreed format (one of the formats could be Oman PINT invoice / Oman TDD) and by internal means of communication and generates:
 - An Oman PINT Invoice
 - An Oman TDD
2. Exchanges in parallel:
 - An Oman PINT Invoice (C2 → C3 flow)
 - An Oman TDD (C2 → C5 flow)
3. Processes the integrated result of the exchanges based on the MLSes received
4. Make all of the PINTs / TDDs available to C1 for auditing and archiving.

Note: Invoice and tax data is stored in transit and then removed, this usage is covered by GDPR (subject to agreement between C1/C2).

5.3 C2 → C3 Flow

Follows the normal Peppol Business Document exchange with the following additions:

- C2 **MUST** generate UUID (see 10.2.3) for the Invoice Business Document (see section 10.2.2) to be included in the PINT Invoice and also the TDD Business Document as a reference to the PINT.
- C2 **MUST** request MLS `ALWAYS_SEND` in the SBDH of the transmission (see Peppol MLS Specification). C2 expects to receive a corresponding MLS in response to transmission of the Invoice Business Document.

The following Invoice Business Document types (see section 10.2.1) **MUST** be exchanged with C3 dependent on C3 registrations in the Dynamic Discovery:

- Oman PINT Billing; Invoice and Credit Note
- Oman PINT Self-Billing, Invoice and Credit Note.

5.3.1 Exceptions (see section 5.6)

1. Negative MLS (Validation error)
 - C3 has (in the MLS) informed C2 that the Invoice Business Document did not validate correctly.
2. Negative MLS (C4 not reachable)
 - C3 has (in the MLS) informed C2 that it could not deliver the Invoice Business Document to C4.
3. MLS not received
 - C2 has not received an MLS from C3 within the timeout defined as SLR [PNP_SLR] in the applicable Operational Guidelines.



5.4 C2 → C5 Flow (same as C3 → C5 flow)

Follows the normal Peppol Business Document exchange with the following additions:

- C2 **MUST** generate UUID (see 10.2.3) and include in the TDD Business Document (see section 10.2.2).
- C2 **MUST** use the UUID determined for the Invoice Business document and include in the TDD Business Document as reference.
- C2 **MUST** request MLS `ALWAYS_SEND` in the SBDH of the transmission (see Peppol MLS Specification) i.e. expects to receive an MLS in response to transmission of the TDD Business Document.

The following TDD Business Document type (see section 10.2.1) **MUST** be exchanged with C5 (see section 10.2.1):

- Oman TDD

5.4.1 Exceptions (see section 5.6)

1. Negative MLS (Validation error)
 - C5 has in the MLS informed C2 that the TDD Business Document did not validate correctly.
2. MLS not received
 - C2 has not received an MLS from C5 within the timeout defined as SLR [PNP_SLR] in the applicable Operational Guidelines.

5.5 C2 - non Peppol domestic

The C1 sender (Oman seller) Participant Identifier Scheme is 0248.

The C4 receiver (Oman buyer non-Peppol) Participant Identifier Scheme is 0248 and the submission will use a “Substitute participant” identifier (see section 10.3.1).

Follows the normal Peppol Invoicing exchange with the following additions/omissions:

- C2 **MUST** have MLS receiving capabilities
- C2 **MUST** generate UUID for the Invoice Business Document (see section 10.2.2)
- C2 **MUST** exchange an OM TDD with C5
- C2 **MUST** not exchange an Invoice Business Document with a C3.

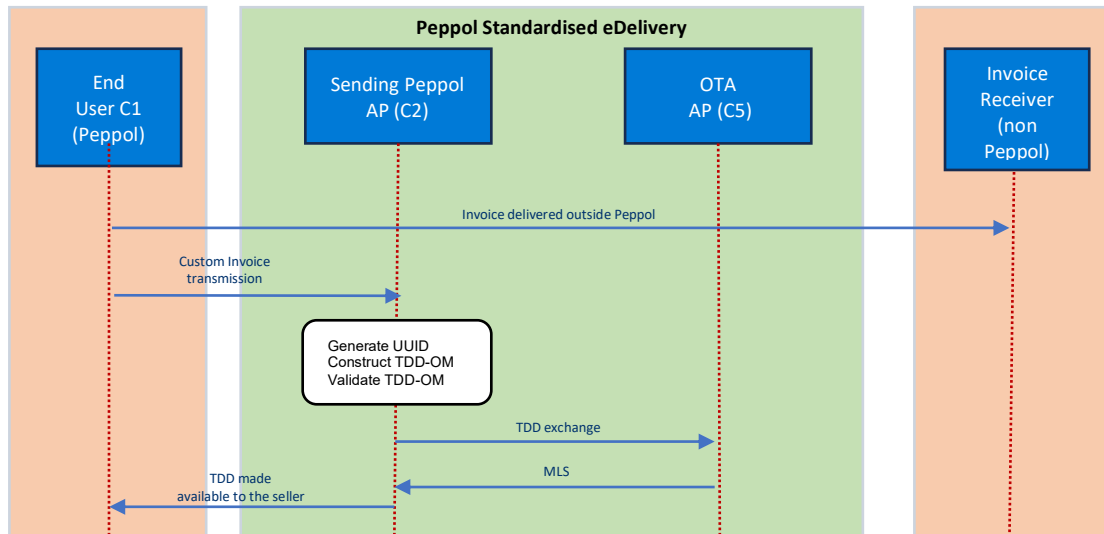


Figure 16 C2 to OM domestic (non Peppol) recipient

5.6 C2 Summary

The following is a summary of all C2 scenarios:

C2 viewpoint	Exchange	Action	Exception
Prerequisite	To SMP	Register: MLS	
Domestic (Peppol)	Prerequisite to sending in parallel to C5 and C3	Generate UUID.	
	To C3	- Use generated UUID. - Exchange Invoice Business Document. - Receive MLS.	Negative MLS (Validation error) - C2 must resolve the exception. - Exchange Invoice Business Document. - Exchange “Disregard” TDD Business Document with C5.
			Negative MLS (C4 not reachable) - Exchange “Disregard” TDD Business Document with C5. - C2 must resolve the exception.
			MLS not received within C3 SLA - C2 must resolve the exception. - Retry Exchange Invoice Business Document.



	To C5	• Use generated UUID. • Exchange “Submit” TDD Business Document. • Receive MLS.	Negative MLS (Validation error) • C2 to resolve the exception. • Exchange “Resubmit” TDD Business Document with C5.
			MLS not received within C5 SLA • C2 to resolve the exception. • C2 check C5 status. • Retry exchange “Submit” TDD Business Document until MLS is Received.
Domestic (non Peppol)	To C5	• Generate UUID. • Exchange TDD Business Document. • Receive MLS.	Negative MLS (Validation error) • C2 to resolve the exception. • Exchange “Resubmit” TDD Business Document with C5.
			MLS not received within C5 SLA • C2 to resolve the exception. • C2 check C5 status. • Retry exchange “Submit” TDD Business Document until MLS is Received.
International (Peppol)	Prerequisite to sending in parallel to C5 and C3	• Generate UUID.	
	To C3	• Use generated UUID. • Exchange International Invoice Business Document. • Receive MLS.	Negative MLS (Validation error) • C2 must resolve the exception. • Exchange Invoice Business Document. • Exchange “Disregard” TDD Business Document with C5.
			Negative MLS (C4 not reachable) • C2 must resolve the exception. • Exchange “Disregard” TDD Business Document with C5.
			MLS not received within SLA • C2 must resolve the exception. • Retry Exchange Invoice Business Document.



	To C5	• Use generated UUID. • Exchange TDD Business Document. • Receive MLS.	Negative MLS (Validation error) • C2 to resolve the exception. Exchange “Resubmit” TDD Business Document with C5.
			MLS not received within C5 SLA • C2 to resolve the exception. • C2 check C5 status. • Retry exchange “Submit” TDD Business Document until MLS is Received.
International (non-Peppol)	Prerequisite to sending in parallel to C5 and C3	• Generate UUID.	
	To C3	• “Substitute participant” identifier used for Receiver (C4) • No exchange.	N/A
	To C5	• Use generate UUID. • Exchange “Submit” TDD Business Document. • Receive MLS.	Negative MLS (Validation error) • C2 to resolve the exception. • Exchange “Resubmit” TDD Business Document.
			MLS not received • C2 to resolve the exception. • C2 check C5 status. • Retry Exchange “Disregard” TDD Business Document until MLS is Received.



6 C3 Capabilities

6.1 C3 Prerequisites

6.1.1 Registration in the Dynamic Discovery

C3 **MUST** be able to support the following Invoice Document Types and register receiving capabilities in the SMP for its C4 identified by:

- Participant Identifier Scheme and Participant Identifier (see section 10.3.1)

With the following Document Types:

- Oman PINT (see section 10.2.1) Invoice – Mandatory receiving capability for Oman (domestic) invoicing organizations
- Oman PINT (see section 10.2.1) Credit Note – Mandatory receiving capability for participating Oman (domestic) invoicing organizations

C3 **MAY** be able to support the following Invoice Document Types and register receiving capabilities in the chosen SMP for its C4 identified by:

- Participant Identifier Scheme and Participant Identifier (see section 10.3.1)

With the following Document Types:

- Oman PINT (see section 10.2.1) Self-billed Invoice – Optional receiving capability for participating invoicing organizations
- Oman PINT (see section 10.2.1) Self-billed Credit Note – Optional receiving capability for participating invoicing organizations

C3 **MUST** register receiving capabilities in the SMP for its own appropriate endpoint by its:

- Service Provider Identification (see section 10.3.3)

With the following Document Type:

- Message Level Status (MLS) [MLS] (see 2.3)

6.1.2 Functional

C3 **MUST** be able to validate (using the Schematrons) the Invoice Business Documents, that it has registered in the Dynamic Discovery on behalf of its End-Users.

C3 **MUST** be able to create an OM TDD from Invoice Business Documents, that it has registered in the Dynamic Discovery on behalf of its End-Users, and exchange with C5.

C3 **MUST** be able to create an MLS from the result of validation of the Invoice Business Documents, that it has registered in the Dynamic Discovery on behalf of its End-Users, and exchange with C2.

C3 **MUST** be able to handle incoming MLS for TDD.

6.2 C3 flow capabilities

C3 follows the flow depicted in the sequence diagram:

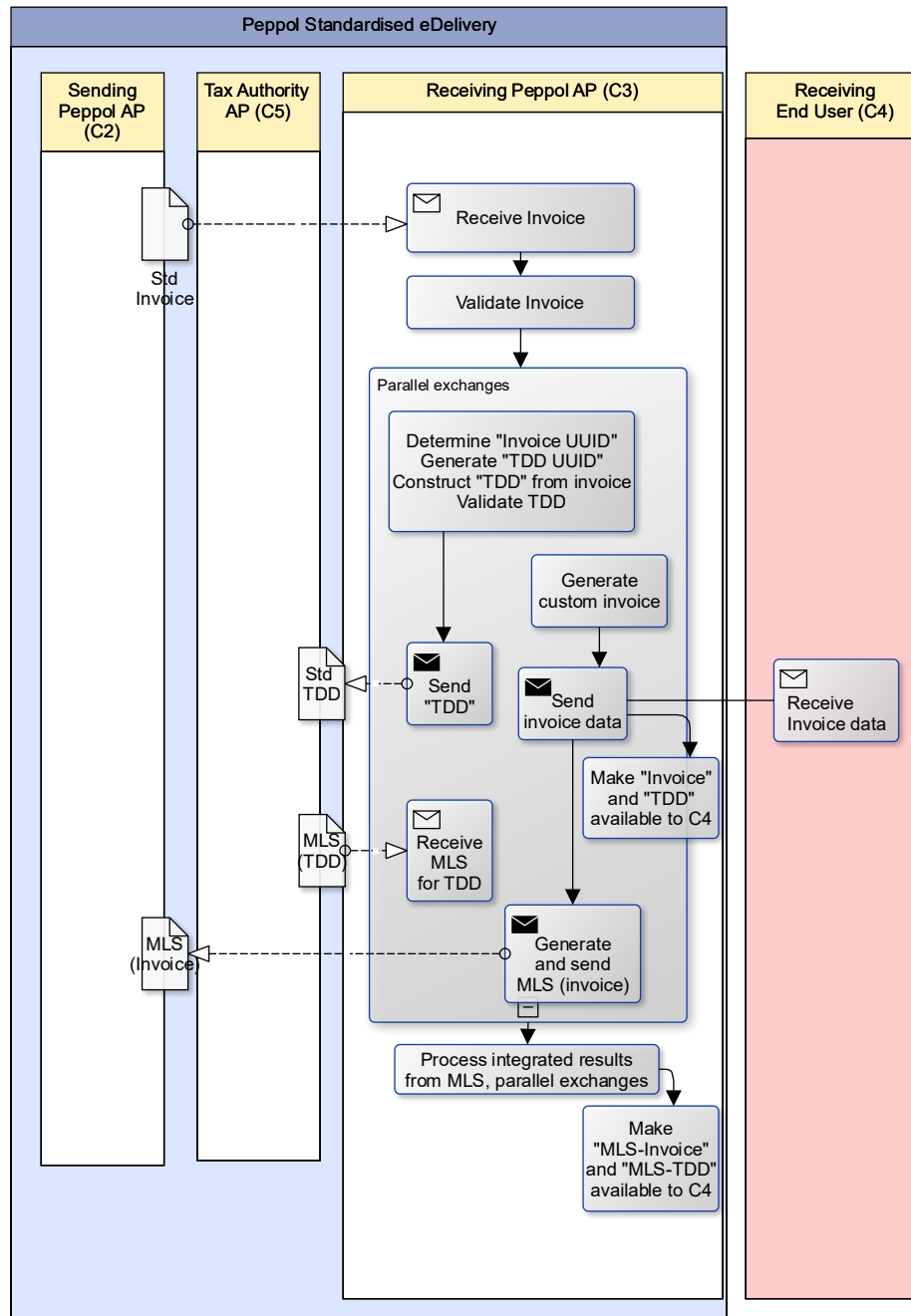


Figure 17: C3 CTC transmissions

The flow depicted in the sequence diagram comprises these steps:

1. C3 receives Oman PINT Invoice in the exchange with C2.
2. C3 Validates the Oman PINT Invoice and generates:
 - An Oman TDD (see note)



- internal (C3-C4) format
- 3. Exchanges in parallel:
 - Invoice in internal (C3-C4) format to C4 by internal means of communication
 - An Oman TDD (C3 → C5 flow) (see note)
 - Generate and send Peppol MLS with (C3 → C2 flow)
- 4. Process the result of the parallel exchanges
- 5. Make all of the Invoices (the original from C2) / TDDs and the Invoice / TDD MLSs available to C4 for auditing and archiving

Note: Invoice and tax data is stored in transit and then removed, this usage is covered by GDPR (subject to agreement between C4/C3).

6.3 C3 → C5 Flow (same as C2 → C5 flow)

Follows the normal Peppol Business Document exchange with the following additions:

- C3 **MUST** generate UUID (see 10.2.3) and include in the TDD Business Document (see section 10.2.2).
- C3 **MUST** use UUID from the Invoice Business Document and include in the TDD Business Document as reference (see section 10.2.2).
- C3 **MUST** request MLS **ALWAYS SEND** in the SBDH of the transmission (see Peppol MLS Specification) i.e. expects to receive an MLS in response to transmission of the TDD Business Document.

The following TDD Business Document type (see section 10.2.1) **MUST** be exchanged with C5 (see section 10.2.1):

- Oman TDD

6.3.1 Exceptions (see section 6.6)

1. Negative MLS (Validation error)
 - C5 has in the MLS informed C3 that the TDD Business Document did not validate correctly.
2. MLS not received
 - C3 has not received an MLS from C5 within the timeout defined as SLR [PNP_SLR] in the applicable Operational Guidelines.

6.4 C3 → C2 Flow

Follows the normal Peppol MLS exchange.



6.4.1 Exception (see section 6.6)

1. Negative Validation of Invoice Business Document
 - C3 validation of the Invoice Business Document did not complete correctly.

6.5 C3/C4 – Peppol/non-Peppol International

The scenario concerns the exchange of invoices where the sending organization is outside Oman and the receiving organisation is within Oman. It covers both Peppol exchange and using other means of exchange e.g. mail.

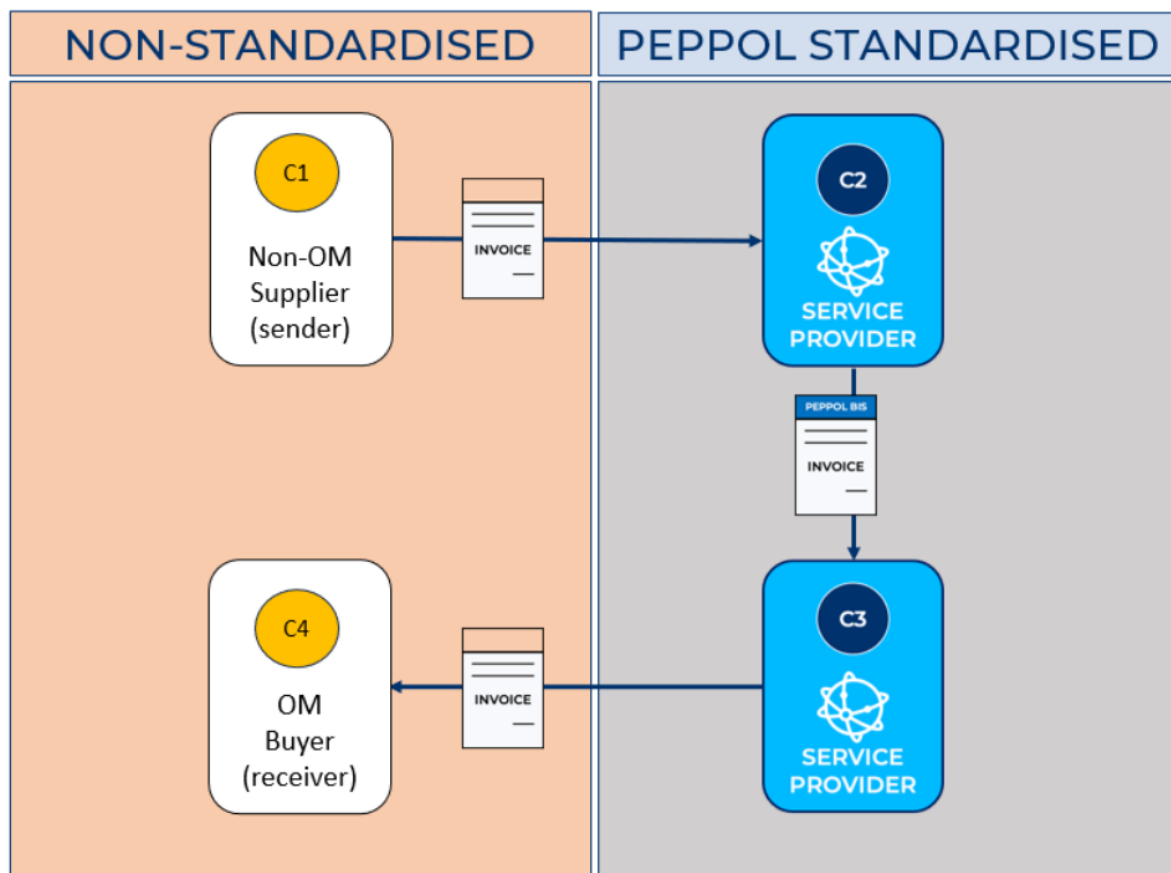


Figure 18: C3 Peppol/non-Peppol International

This scenario does not result in a TDD produced by Oman Invoice Receivers, because it cannot be guaranteed that the Invoice can provide all required data necessary for creating an OM TDD. However, in this case the Buyer will issue a self-billed Invoice for tax reporting purposes.

6.6 C3 Summary

The following is a summary of C3 scenarios:



C3 viewpoint	Exchange	Action	Exception
Prerequisite	To SMP	Register: - Invoice Business Documents - MLS	
Domestic	From/To C2	- Receive Invoice Business Document. - Validate exchanged Invoice Business Document. - Exchange MLS.	Validation error - Exchange negative MLS with C2. - Note: No TDD is exchanged with C5
			C4 not reachable - Exchange negative MLS with C2 - Note: No TDD is exchanged with C5
	To C5	- Exchange “Submit” TDD Business Document. - Receive MLS.	Negative MLS (Validation error) - C3 to resolve the exception. - Exchange “Resubmit” TDD Business Document with C5.
			MLS not received within C5 SLA - C3 to resolve the exception. - C3 check C5 status. - Retry exchange “Submit” TDD Business Document until MLS is Received.
Domestic (non Peppol)	To C5	- Exchange “Submit” TDD Business Document. - Receive MLS.	Negative MLS (Validation error) - C3 to resolve the exception. - Exchange “Resubmit” TDD Business Document with C5.
			MLS not received within C5 SLA - C3 to resolve the exception. - C3 check C5 status. - Retry exchange “Submit” TDD Business Document until MLS is Received.
International (Peppol)	From/To C2	- Receive Invoice Business Document. - Validate exchanged Invoice Business Document. - Exchange MLS.	N/A



	To C5	No Exchange.	N/A
International (non-Peppol)	To C5	No Exchange.	N/A
	To C2	No exchange.	N/A



7 C5 Capabilities

7.1 C5 Prerequisites

7.1.1 Registration in the Dynamic Discovery

C5 **MUST** register receiving capabilities in the SMP for the OTA (C6) identified by:

- OTA (C6) Identifier (see section 10.3.2)

With the following Document Types:

- Oman Tax Data Document (see section 10.2.1)

Note: C5 must be registered in the Oman SMP as an AP with one customer (C6).

7.1.2 Functional

C5 **MUST** be able to validate the Oman TDD Business Document (using the Oman TDD Schematron - and not for Tax compliance etc.).

C5 **MUST** be able to create an MLS from the result of validation of the TDD Business Document and exchange with C2 or C3.



7.2 C5 flow capabilities

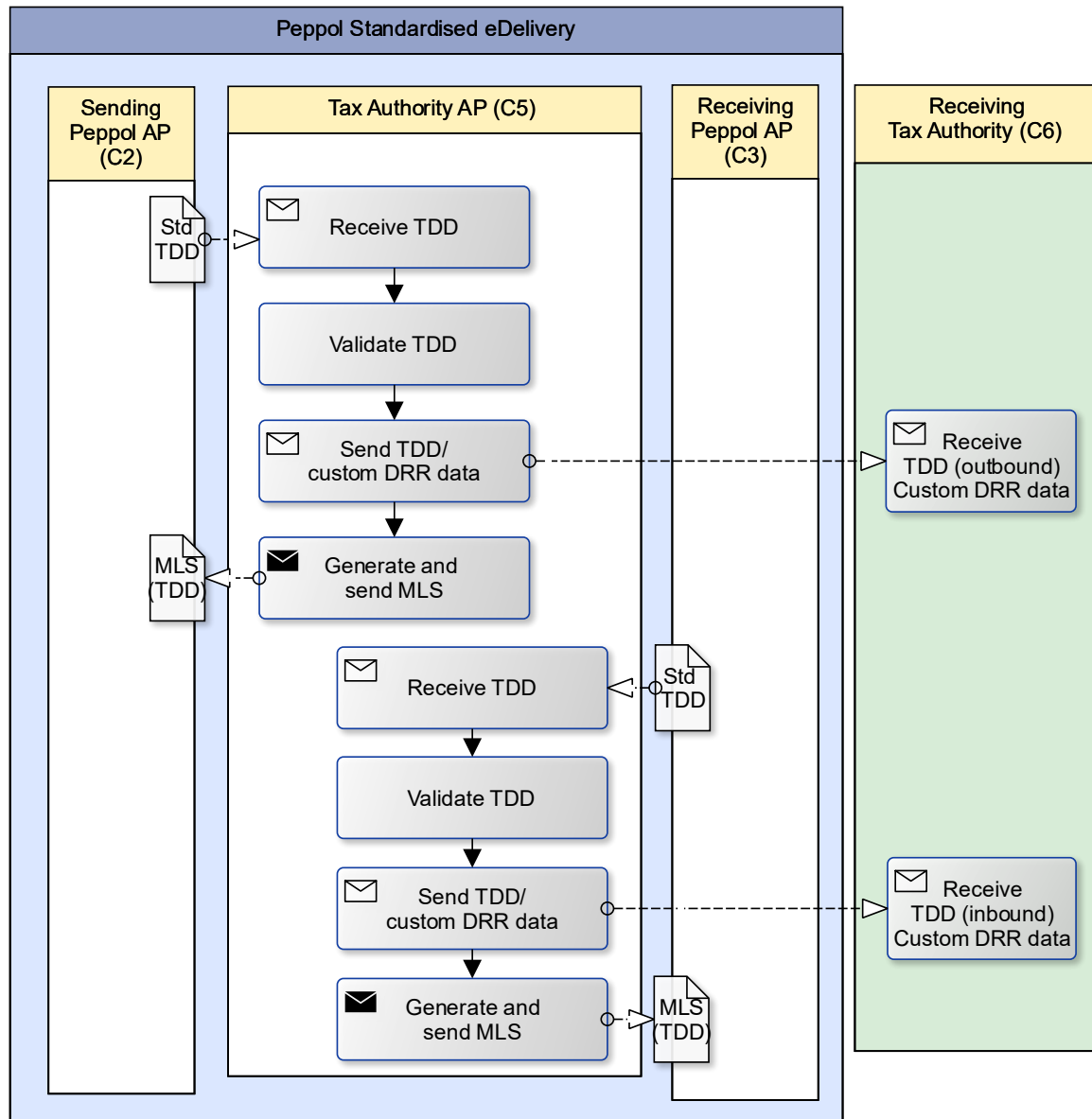


Figure 19: C5 CTC transmissions

C5 follows the flow depicted in the sequence diagram:

1. C5 receives Oman TDD in the exchange with C2 or C3.
2. C5 Validates the Oman TDD and transmit to C6 the Tax Data in internal (C5-C6) format and by internal means of communication, and generates:
 - a. A Peppol MLS
3. Exchanges:
 - a. A Peppol MLS (C5 → C2/C3 flow)



7.3 C5 → C2/C3 Flow

Follows the normal Peppol MLS exchange.

7.3.1 Exception (see section 7.4)

1. Negative Validation of TDD Business Document
 - The validation of the TDD Business Document did not validate correctly.

7.4 C5 Summary

The following is a summary of C5 scenarios:

C5 viewpoint	Exchange	Action	Exception
Prerequisite	To SMP	Register: TDD Business Document	
Domestic International (Peppol) International (Non-Peppol)	From/To C2	- Receive TDD Business Document. - Validate exchanged TDD Business Document. - Exchange MLS.	Validation error Exchange negative MLS with C2 or C3. Abort Processing.
	From/To C3	- Receive TDD Business Document. - Validate exchanged C3 to C5 TDD Business Document. - Exchange MLS.	Validation error Exchange negative MLS with C3. Abort Processing.



8 C6 TDD submission statuses

The following model outlines the Tax Data Document (TDD) status transitions applicable to the OTA (C6).

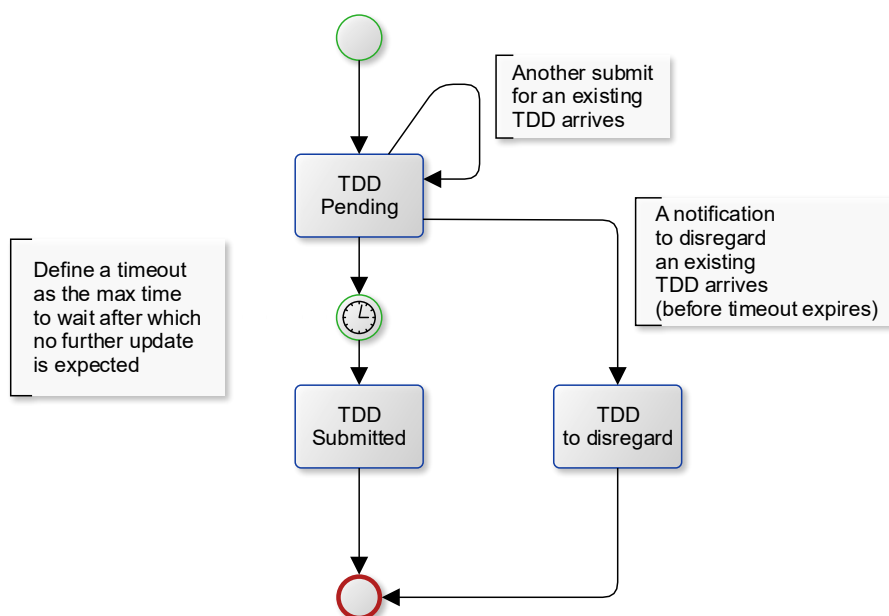


Figure 20: C6 TDD submission notifications

8.1.1 Normal

Table 1 C6 Normal consolidation table

TDT006 received	Situation	action
Submit	No other TDD has been received on this Invoice	Use TDD
Resubmit	A previous TDD has been received on this TDD. i.e. a new TDD has been received to replace an old TDD	Mark old (previously received) TDD as "Disregard" and use new TDD as current instead
Disregard	A previous TDD has been received on this TDD. i.e. an older TDD has been received against an existing newer TDD (potentially due to a delay)	Mark the older (but most recently received) TDD as "Disregard"



8.1.2 Exception

Table 2 C6 Exception consolidation table

TDT006 received	Situation	action
Submit	A previous TDD has been received on this TDD	Replace old TDD with new TDD (Treat as “Resubmit”)
Resubmit	No other TDD has been received on this Invoice (Exception of C2/C3)	No previous TDD to replace (Treat as “Submit”)
Disregard	A previous TDD has not been received on this TDD (Exception of C2/C3)	No action



9 Oman document flows

The flow of all documents in the Peppol network are shown for Oman:

- Domestic B2B / Domestic B2C
- import / export
- Profit margin

Table 3 OM Peppol document flows

ID	Jurisdiction	Scenario	document flows
1	OM domestic B2B	Seller and Buyer both VAT registered and part of the eInvoicing network in Oman	C2->C3 Invoice C2->C5 TDD-OM C3->C5 TDD-OM C3->C2 MLS C5->C2 MLS C5->C3 MLS
2	OM domestic B2B	Only seller is VAT registered. Buyer is not VAT registered hence cannot be part of the eInvoicing network	C2->C5 TDD-OM C5->C2 MLS
3	OM domestic B2B	Only buyer is VAT registered (for example in case of Profit margin self-invoice)	C3->C5 TDD-OM C5->C3 MLS
4	OM domestic B2B	Only seller is registered in Oman SMP and part of the eInvoicing network currently. The buyer (although VAT registered) is not part of the eInvoicing implementation roll out and hence not on the Oman SMP	C2->C5 TDD-OM C5->C2 MLS
5	OM domestic B2B	C4 not reachable (but registered in Oman SMP)	C2->C3 Invoice C2->C5 TDD-OM



			C5->C2 MLS C3->C2 MLS (negative) C2->C5 TDD-OM(D)
6	Profit margin	Only seller is VAT registered OR both seller and buyer are VAT registered	C2->C5 TDD-OM C5->C2 MLS
7	OM domestic B2B – Self Billed	Seller and Buyer both VAT registered and part of the eInvoicing network in Oman, but the invoice is issued by the buyer (C4) on behalf of the seller (C1)	C3->C2 Invoice C2->C5 TDD-OM C3->C5 TDD-OM C3->C2 MLS C5->C2 MLS C5->C3 MLS
8	OM domestic B2B – Third Party	Seller and buyer are both VAT registered however the seller appoints a VAT registered third party to issue invoices on behalf of the seller. In this case the third party is considered to be C1. Exchange of invoices between the seller and third party must be managed between the seller and the third party.	C2->C3 Invoice C2->C5 TDD-OM C3->C5 TDD-OM C3->C2 MLS C5->C2 MLS C5->C3 MLS
9	OM domestic B2C	Only seller is VAT registered	C2->C5 TDD-OM C5->C2 MLS
10	OM domestic B2C – Third Party	Only seller is VAT registered however the seller appoints a VAT registered third party to issue invoices on behalf of the seller. In this case the third party is considered to be C1. Exchange of invoices between the seller and third party must be	C2->C5 TDD-OM C5->C2 MLS



		managed between the seller and the third party.	
11	Export	outbound invoice (Seller is VAT registered in Oman and Buyer is outside Oman)	C2->C5 TDD-OM C5->C2 MLS
12	Import	inbound invoice (Buyer is VAT registered in Oman and Seller is outside Oman)	C3->C5 TDD-OM C5->C3 MLS



10 Identifiers and Specifications

10.1 Process Identifiers

Process is the sequence of Document Types being exchanged between stakeholders and therefore define the Choreography. In OM Peppol Invoicing Process the exchange choreographies are:

- C1 → C2 → C3 → C4 (Invoice)
- (C1) → C2 → C5 → OTA (TDD)
- (C4) → C3 → C5 → OTA (TDD)

“Peppol Policy for use of Identifiers” section 6 describes the policies for Process Identifiers [eDN].

Peppol Invoice / TDD Process Identifier Scheme: `cenbii-procid-ubl`

Peppol Invoice Process Identifier Value:

- `urn:peppol:bis:billing`
- `urn:peppol:bis:selfbilling`

Peppol TDD Process Identifier Value:

- `urn:peppol:taxreporting`

All Process Identifications can be found at: <https://docs.peppol.eu/edelivery/codelists/> (Processes).

10.2 Documents identifiers and Specifications

10.2.1 Invoice Business Document Types

Peppol Invoice Business Document Type

The allowed Invoice Business Documents are:

- Oman PINT – includes document types for Invoice and Credit Note
- Oman PINT – includes document types for Self-Billed Invoice and Credit Note

Peppol Document Type Identifier Scheme: `peppol-doctype-wildcard`

Peppol Document Type Identifiers (proposed):

`urn:oasis:names:specification:ubl:schema:xsd:Invoice-2::Invoice##urn:peppol:pint:billing-1@om-1::2.1`

`urn:oasis:names:specification:ubl:schema:xsd:CreditNote-2::CreditNote##urn:peppol:pint:billing-1@om-1::2.1`



```
urn:oasis:names:specification:ubl:schema:xsd:Invoice-2::Invoice##urn:peppol:pint:selfbilling-1@om-1::2.1
```

```
urn:oasis:names:specification:ubl:schema:xsd:CreditNote-2::CreditNote##urn:peppol:pint:selfbilling-1@om-1::2.1
```

Specification: **Peppol OM PINT Document:**

```
https://test-docs.peppol.eu/pint/pint-om/
```

10.2.2 TDD Business Document Type

Peppol Tax Data Document

Peppol Document Type Identifier Scheme: `busdox-docid-gns`

Peppol Document Type Identifier: `urn:peppol:schema:om-taxdata:1.0::TaxData##urn:peppol:taxdata:om-1::1.0`

Specification: **Peppol OM Tax Data Document:**

```
https://test-docs.peppol.eu/pint/pint-om/om-tdd/
```

10.2.3 Business Document and Transmission instances

For traceability and reconciliation of both Invoice and TDD Business Documents, a UUID **MUST** be used.

Invoice document unique identifier

Invoice documents are referenced by a UUID that uniquely identifies it. UUID version 5 [UUID]¹ is used. This means that the same content invoice has the same invoice UUID, so if the content changes a new UUID is determined. The Invoice UUID is determined by C2 and C3 in the same way. Common programming languages (like Java, Python and C#) have mature libraries for generation of UUID Version 5.

The Invoice UUID is calculated based on a “name” containing values of these invoice fields:

- IBT-34-1 (attribute) scheme ID (mandatory)
- IBT-34 seller identifier ID (mandatory)
- IBT-003 invoice type code (mandatory)
- IBT-001 invoice number (mandatory)
- IBT-002 invoice date (mandatory)
- IBT-110 invoice total tax amount (mandatory)

¹ UUID v5 is a standard for a hash-based, deterministic identifier based on specified payload data (see <https://www.rfc-editor.org/rfc/rfc9562#name-uuid-version-5>)



- IBT-112 invoice total amount with TAX (mandatory)
- BTOM-020 Total amount due (Profit Margin invoices only)

The **Invoice UUID** (and **Seller UUID** – see section 4.2) **MUST** use this fixed namespace²:

`e0bc4ac8-b025-46e5-a76d-0c893fc3027e`

Notes:

- The above fields are all mandatory in the invoice, so the invoice will have already been rejected if any of the above are missing, in this case no TDD / UUIDv5 will be created.
- All leading and trailing whitespace should be removed from above values in the invoice
 - White space consists of one or more of the following characters: space (#x20), tab (#x9), carriage return (#xD), or line feed (#xA).
- In the input string each value is separated with a single blank character (' ', ASCII 32)
- Date format YYYY-MM-DD
 - The date value **MUST** be formatted according to the XML Schema date format rules and **MUST NOT** contain time zone information
- The UUIDv5 'name' should use the UTF-8 character set.

An example UUIDv5:

- input string: `0088 5060012349998 380 33445566 2026-01-13`
- output string: `1780de4f-a87c-50cc-9d8a-f982abe36912`

As the Invoice UUID can be easily determined from any instance, it is not required to be included in the instance itself.

- For example, in an example where C3 only submits an OM TDD, and no invoice was received from C2, C3 will be able to create the correct Invoice UUID when it is not provided in the PINT.

Tax Data document unique identifier

TDD documents **MUST** contain a UUID that uniquely identifies it. UUID version 4 [UUID] is used and generated locally by C2 and C3. Common programming language (like Java, Python, C#) have mature libraries for generation of UUID Version 4.

The TDD UUID is placed in the TDD Business Document (TDT-003, Unique Identifier Number).

² Generated UUIDv4 per the recommendation of RFC9562
<https://www.rfc-editor.org/rfc/rfc9562#name-namespace-id-usage-and-allo>



UUID MUST be represented in hexadecimal value e.g.

bfeac3d0-82d8-4195-9d88-53b3547f83a

Note that also the SBDH uses UUID (v4) in element

`DocumentIdentification/InstanceIdentifier` to identify the transmission within the Peppol Network.

For traceability of Invoice Business Documents, it must be possible to reference both the Business Document and the Transmission.

- The Transmission is identified by the UUID in the SBDH (see 11.1)
- The Business Document is identified by the Invoice UUID of the Business Document
- The Tax data document is identified by the TDD UUID of the Business Document
- The B2C receipt is identified by the Seller UUID included in the QR code on the (readable) invoice

The following model shows the identification and references to identifications in the different Business Documents and MLS:

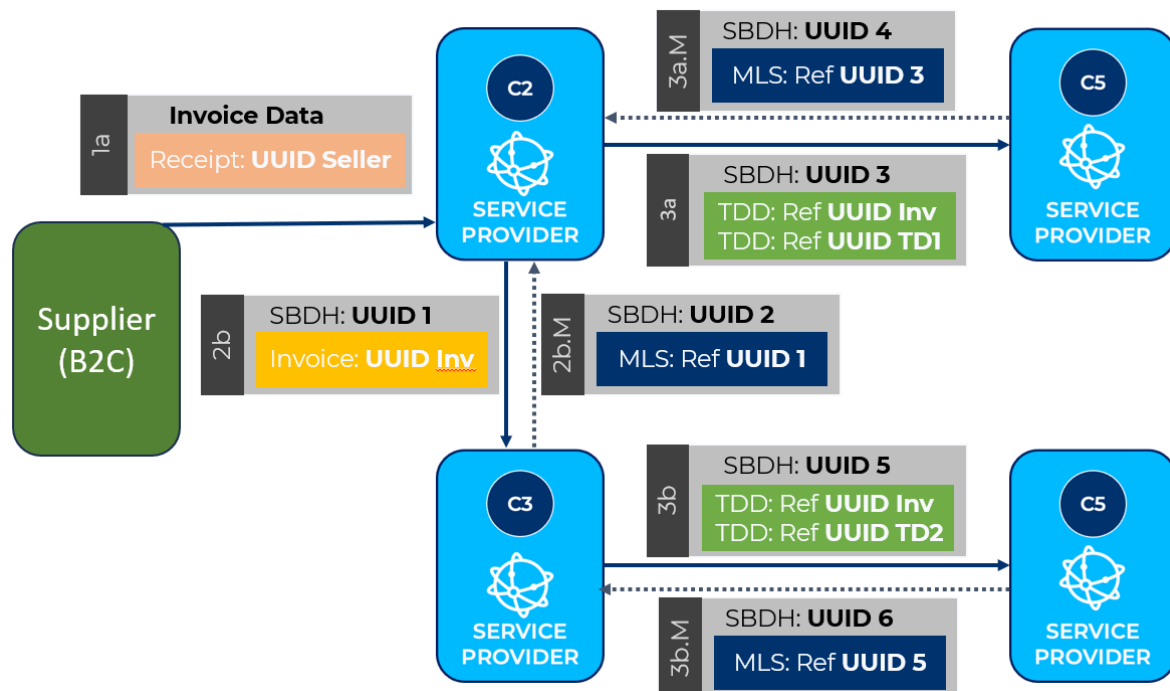


Figure 21: Identification and references from other Business Documents

- B2C receipt (1a) is determined by the supplier.
- The transmission of the Invoice Business Document identifications (2a) is determined by C2.



- Both, the transmission and the Invoice Business Document are referenced in the TDDs from C2 and C3 (3a and 3b)
- An MLS references the transmission identification of the validated Business Document (3a.M, 2b.M, 3b.M).

UUID fields in the Invoice, TDD and BC2 receipt.

Table 4 UUID and business documents table

UUID	UUID Version	Generated by	PINT Invoice	OM TDD	Receipt
Seller UUID (1a)	UUID Version 5	Seller	BTOM-004	BTOM-004	In QR code
Invoice UUID (2b)	UUID Version 5	C2	BTOM-002 Also: • BTOM-031 • BTOM-004 • BTOM-023	BTOM-002	No
TDD UUID(s) (3a, 3b)	UUID Version 4	C2, C3	No	TDT-003	No
Transmission UUID (2b.M, 3a.M, 3b.M)	UUID Version 4	C2, C3, C5	No	No	No

10.3 Identification of Participants

10.3.1 End-Users (C1, C4)

The principal basis of identifying End-Users i.e. C1 (sender) and C4 (receiver) Oman participants (organisations):

The usage of the identifier is:

- Inside the Business Documents.
- In the SBDH.



- In the SMP.

Oman issues a VAT Identification Number (12 digits) to organisation that are:

- Oman VAT Registrants

For example; 0248:OM1100003554.

These can act as C1 or C4, with a Participant Identifier Scheme (0248) in the Peppol Participant Identifier Scheme code list [CL].

“Substitute participant ID” identifies domestic / international invoice receivers in non-Peppol Invoice exchanges that are not registered in Peppol (the actual identification of the receiver must be completed by other means).

No.	Transaction type	Participant scheme ID	Participant Identification
1	When the invoice transaction type code is (XXXXXXXX1XXXXXXXXXXXXX) Deemed supply (B2B transactions only) and if invoice must not be shared with the buyer Note: If the invoice is a deemed supply for B2C transactions then fixed value for B2C transactions must be used	0248	997770000096
2	When the invoice transaction type code is (XXXXXX1XXXXXXXXXXXXX) Exports (receiver not registered in Peppol)	0248	997770000097
3	When the Self Billing Invoice (import) is not subject to Oman eInvoicing regulations	0248	997770000098
4	Where the invoice relates to a B2C transaction (domestic) and/or relates to profit margin scheme (XXXXXXXX1XXXXXXXXXXXXX)	0248	997770000099
5	Where the buyer (domestic) is not part of the Peppol network (excluding. B2C) during partial roll out	0248	997770000095



10.3.2 Tax Authorities (C6)

Identifying C6 (OTA)

For the purpose of identifying the OTA, use the Service Provider Identifier Scheme (SPIS) with the ICD code 0242. The Identifier is divided into three parts:

- Service Provider Identifier: the 6 digit “Seat Id” from the Peppol AP Certificate (CN)
 - For example (OM) – 001090
- Use Case: TaxAuthority
- Suffix: “OM”

Example: 001090-TaxAuthority.OM

Note: C1/C2 submits the TDD to C6 (OTA), via the C5 (C6's AP), C5 registers the receiving capabilities of the C6 (see section 7). C5 then delivers the TDD to C6 in the agreed format, this final delivery is outside the Peppol eDelivery network (like C3 delivering invoice data to C4) – see Figure 4: Full model with exchange from C1 to C6.

10.3.3 Service Providers / Access Points (C2, C3, C5)

Service Provider Identifier Scheme (SPIS) with the ICD code 0242.

For example; 0242:123456 (where the SPIS is combined with the SP seat ID).



11 Transmission

11.1 SBDH Setup

The transmission of Business Document and MLS messages is performed using Peppol eDelivery AS4.

The Business Document is enveloped in a Peppol SBDH envelope (Standard Business Document Header), which holds the metadata of the transmission e.g. Sender Identifier (C1), Receiver Identifier (C4), Process Identifier, Document Type Identifier and the Country Code of the Sender [eDN].

SBDH XML Invoice

```

<StandardBusinessDocument
  xmlns="http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader">
  <StandardBusinessDocumentHeader>
    <HeaderVersion>1.0</HeaderVersion>
    <Sender>
      <Identifier Authority="iso6523-actorid-upis">C1_SenderId</Identifier> 3
    </Sender>
    <Receiver>
      <Identifier Authority="iso6523-actorid-upis">C2_ReceiverId</Identifier> 4
    </Receiver>
    <DocumentIdentification>
      <Standard>Namespace_of_document_type_standardization</Standard>
      <TypeVersion>document_type_version</TypeVersion> 2
      <InstanceIdentifier>4d386b06-f4a3-4d74-bfbb-31668bd153b7</InstanceIdentifier> 1
      <Type>Type_of_Document</Type>
      <CreationDateAndTime>YYYY-MM-DDTHH:MM:SSZ</CreationDateAndTime>
    </DocumentIdentification>
    <BusinessScope>
      <Scope>
        <Type>MLS_TO</Type>
        <InstanceIdentifier>C2_ReceiverId</InstanceIdentifier>
      </Scope>
      <Scope>
        <Type>MLS_TYPE</Type>
        <InstanceIdentifier>MLS_Handling_Code</InstanceIdentifier> 5
      </Scope>
      <Scope>
        <Type>DOCUMENTID</Type>
        <InstanceIdentifier>Document_Type_Identifier</InstanceIdentifier> 6
        <Identifier>busdox-docid-qns</Identifier>
      </Scope>
      <Scope>
        <Type>PROCESSID</Type>
        <InstanceIdentifier>Process_Identifier</InstanceIdentifier> 7
        <Identifier>cenbii-procid-ubl</Identifier>
      </Scope>
      <Scope>
        <Type>COUNTRY_C1</Type>
        <InstanceIdentifier>Country_of_C1</InstanceIdentifier> 8
      </Scope>
    </BusinessScope>
  </StandardBusinessDocumentHeader>
  <DocType>
    xmlns:cbc="urn:oasis:names:specification:ubl:schema:xsd:CommonBasicComponents-2"
    xmlns:cac="urn:oasis:names:specification:ubl:schema:xsd:CommonAggregateComponents-2"
    xmlns="urn:oasis:names:specification:ubl:schema:xsd:Invoice-2">
      <!-- ("Invoice" or TDD Business Document) -->
    </DocType>
  </StandardBusinessDocument>

```

Figure 22: SBDH XML



The following restrictions apply to the attributes in the Peppol SBDH specification for C2 to C3 transmission of Invoice Business Documents on behalf of a C1:

1. UUID version 4 generated by the Sender AP, uniquely identifying the transmission, see section 10.2.2
2. SBDH follows the general rules for Business Document payloads i.e. UBL 2.1 content is indicated by `<TypeVersion>2.1</TypeVersion>`
3. For a C1, `Sender/Identifier` MUST be of Participant Identifier Schemes associated with the taxable persons, see section 10.3.1
4. For a C4, `Receiver/Identifier` MUST be of Participant Identifier Schemes associated with the taxable person, see section 10.3.1
5. For `MLS_TYPE`, MUST have `ALWAYS_SEND`
6. For `DOCUMENTID`, the `Document_Type_Identifier` MUST be one of the identifier values stated in section 10.2
7. For `PROCESSID`, the `Process_Identifier` MUST be one of the values for the process identifiers as stated in section 10.1
8. For `COUNTRY_C1`, MUST have the value as appropriate for the sending End User as required for Peppol traffic reporting.



12 Security

12.1 Peppol Trust Model

The Peppol Trust model is used for all three different flows:

1. Invoice exchange: C1 → C2 → C3 → C4 involving an Invoice Business Documents
2. Tax Data exchange: (C1 →) C2 → C5 → C6 involving a Tax Data Document
3. Tax Data exchange: (C4 ←) C3 → C5 → C6 involving a Tax Data Document

In the Peppol Post-Award domain a Peppol Post-Award Certificate for the invoice exchange is used for both Authenticity and Authorization. This also applies to the two Tax Data exchanges.

12.2 C5 segregation

In the 5-corner/6-corner model, the C5 Access Point acts on behalf of the Tax Authority (C6). Therefore, the C5 service provider must either:

- Be organizationally separate from any C2/C3 service provider roles, or
- Logically segregate its C5 Access Point functions from any other Access Point services (e.g., for C2 or C3).

12.3 SMP and Portal systems

The Oman solution includes a centralised SMP and also a tax payer / SP portal, which will manage SMP updates. It is important that the SMP and the Portal system(s) have clearly defined segregation.

The portal should not update the SMP directly, but manage the process of update requests and approval.



Appendix A References

The references are used throughout the document where applicable, often augmented with a specific reference within the quoted document like a legal article or a specific section or page.

Reference	Full name and document version info
[EN16931]	European Norm on Electronic Invoicing in public procurement Electronic Invoicing – Part 1: Semantic data model of the core elements of an electronic invoice Ref. No. EN 16931-1:2017+A1:2019
[TASQ-Sec xx]	Structured Questionnaire to Tax Administrations V1.1 of 17.12.2024 (referencing specific sections by heading number)
[PIF]	Peppol Interoperability Framework (see https://peppol.org/learn-more/peppol-interoperability-framework/) Including all its constituent specifications in the “Peppol Architectural Framework”
[P-CTC-Ref]	Peppol CTC Reference Document V1.0, Sept. 2021 https://peppol.org/wp-content/uploads/2023/02/Peppol-CTC-Reference-Document-v1.0.pdf
[P-CTC-RefAdd]	Peppol CTC Reference Document Addendum V0.7, Oct. 2010 https://peppol.org/wp-content/uploads/2023/10/Peppol-CTC-Reference-Document-Addendum-v1.07.pdf
[DCTCE]	A Next-Generation Model for Electronic Tax Reporting and Invoicing - DCTCE GENA, v2.0 of 3.8.2022 https://www.gena.net/l/library/download/urn:uuid:c643584f-5fed-4038-9d51-52aeb1e4b19a/next+generation+model+-+decentralised+ctc+and+exchange+v2.0.pdf
[UUID]	RFC 9562 – Universally Unique Identifiers (UUIDs) IETF, May 2024 (obsoletes RFC 4122) https://www.rfc-editor.org/rfc/rfc9562



[PeNCA]	Peppol eDelivery Network Conceptual Architecture https://openpeppol.atlassian.net/wiki/
[PeNLA]	Peppol eDelivery Network Logical Architecture https://openpeppol.atlassian.net/wiki/
[MLS]	Peppol Message Level Status Specification https://docs.peppol.eu/edelivery/specs/mls/v1.0.0/
[PNP_SLR]	Peppol Network Policy 1.0.0-RC 2025-12-01 (includes SLR for MLS)
[SMP]	Peppol Service Metadata Publishing https://docs.peppol.eu/edelivery/smp/Peppol-EDN-Service-Metadata-Publishing-1.4.0-2025-02-06.pdf
[SML]	Service Metadata Locator (SML) https://docs.peppol.eu/edelivery/sml/Peppol-EDN-Service-Metadata-Locator-1.3.0-2025-02-06.pdf



Appendix B Terminology

To ease wording in the body of this document, it is useful to define a few concepts as “fixed terms” to reduce ambiguity and increase conciseness of the content. Wherever possible, existing terminology is used and imported by (brief explanatory) reference.

To indicate intentional use of fixed terms, they are spelled using **Upper Case** notation.

<i>Term</i>	<i>Description</i>
Cross-Border Invoicing	denotes invoicing transactions where the issuer and recipient fall under 2 different jurisdictions. This includes all cases irrespective of whether the one or both jurisdictions are an EU member state, i.e. intra-community and 3 rd -country import/export scenarios.
Peppol Interoperability Framework (PIF)	The set of artifacts (i.e., agreements, policies, procedures and technical specifications) which together ensure interoperability in the Peppol Network. It consists of the Peppol Architectural Framework and the Peppol Governance Framework and evolves according to the change management provisions set forth in the Internal Regulations and the Operational Procedures and the principles set out in this agreement.
Access Point (AP)	denotes the services including corresponding organizational units directly interfacing in the Peppol network with other Access Points. An Access Point is a specific part of a Service Provider, which may include more services additional to those directly related to Peppol network operations.
Service Provider (SP)	Any accredited network party assuming a C2/C3 role in the Peppol framework or equivalent functions in other network models. An organisation authorised to provide Peppol Services within one or more Peppol Service Domains pursuant to a Peppol Service Provider Agreement.
End User	Any business-level party assuming a C1/C4 role, i.e. contracting (directly or indirectly) a Service Provider to implement business message flows with their trading partners. An identified or identifiable entity that is responsible for the business content of the datasets that is exchanged (by sending and/or receiving) with another such entity using Peppol Services over the Peppol Network.
SP Customer	The business-level entity contracting a Service Provider to implement their business flows with their Trading Partners, In the context of B2B Interoperability, the “End User” is always an “SP Customer” by definition – otherwise it wouldn’t be “interoperability”.



Continuous Transaction Controls (CTC)	denote a tax reporting approach based on near-time, continuous collection of financial transaction data, as opposed to relying on summarised periodic reports.
Digital Reporting Requirements (DRR)	denotes one of the three pillars in the “VAT in the Digital Age” Council Directive, specifying the legal regulations towards an EU-wide reporting system on Intra-Community business-to-business/-government transactions.
Tax Reporting	is the process of extracting, submitting and evaluating tax data for the sake of fulfilling tax obligations. Generally, in the context of Tax reporting enabled countries, a real-time reporting process is assumed, as opposed to periodic/aggregated reporting.
Real-Time Tax Reporting	where required to distinguish from “periodic” tax reporting for clarity, explicitly spells out the “real-time” character of the tax reporting process in question.
Tax Data Document (TDD)	Electronic document carrying tax data for reporting to tax authorities from businesses.
Core Invoice Usage Specification (CIUS)	is a restriction or extension of the core invoice model to meet national or sector-specific requirements. As defined in EN 16931.
PASR	Peppol Authority Specific Requirements.



Appendix C SLR/SLR Timings

The standard Peppol SLR timings for MLS are defined in the SLR [PNP_SLR]. Oman will align to these SLR timings.

SLR MLS-1 – Latest MLS Sending

The sending of an MLS message is constraint by the following rule with a monthly measurement period:

- 99.5% of all MLS messages for business documents with a Payload Size of less than 10 Megabytes MUST have a maximum duration between MLS Milestone M1 and MLS Milestone M2.

SLR MLS-2 – Latest MLS Reception

The receiving of an MLS message is constraint by the following rule with a monthly measurement period:

- 99.5% of all MLS messages for business documents with a Payload Size of less than 10 Megabytes MUST have a maximum duration between MLS Milestone M1 and MLS Milestone M3.

Definitions:

- MLS Milestone 1 (M1) - The MLS milestone M1 is the date and time when the transmission of the Peppol Dataset is initiated.
- MLS Milestone 2 (M2) - The MLS milestone M2 is the date and time when the transmission of the MLS message is initiated.
- MLS Milestone 3 (M3) - The MLS milestone M3 is the date and time when the MLS message is received on the Access Point.

Notes:

1. C5 should be treated as another access point until the [PNP_SLR] is updated to included C5s
2. The Payload Size MUST be measured as the size of the uncompressed, unencrypted and unsigned Peppol Business Message Envelope instance in bytes, excluding any present leading and trailing whitespace characters.

SLA times for Oman will be covered by local legislation

For testing invoice and TDD submission the below times will be configured in the testbed. This is for the purpose of testing within a limited timeframe, but allows SPs to demonstrate they can manage to comply with a set SLA time (which could vary in production).

Note: For production Oman will align to PNP SLRs for MLS, and for TDDs the SLAs are specified in the Oman PASR.



Oman Testing timings (only):

Scenario	SLA time
The maximum time period within which an MLS message MUST be sent by C3 or C5 and received by C2. The maximum time shall be calculated between the reference points of: - The time when the original business document was sent by C2 - The time when the MLS has been received by C2 <i>Note: this MLS timing applies to testbed only, for production Oman will align with the MLS SLR from the [PNP_SLR]</i>	20 mins
The maximum time period for C2 to send the TDD to the Peppol Authority of Oman (C5) for B2B/G2B transactions. The maximum time shall be calculated between the following reference points: - The time when the original business document is validated by C2 - The time when the TDD is transmitted to C5 <i>Note: for production timings see the PASR</i>	15 mins
The maximum time period for C2 to send the TDD to the Peppol Authority of Oman (C5) for B2C/G2C transactions. The maximum time shall be calculated between the following reference points: - The time when the original business document is validated by C2 - The time when the TDD is transmitted to C5 <i>Note: for production timings see the PASR</i>	30 mins
The maximum time period for C3 to send the TDD to the Peppol Authority of Oman (C5) for B2B/G2B transactions. The maximum time shall be calculated between the following reference points: - The time when the original business document is received by C3 - The time when the TDD is transmitted to C5 <i>Note: for production timings see the PASR</i>	15 mins



C.1 Retention policy

Scenario	Retention time
Accredited Service Providers shall retain transmission and transaction metadata fields related to Electronic Tax Invoices and Electronic Adjustment Notes processed through their systems, which shall include at a minimum: • Sender and receiver identifiers • Transmission timestamps • Acknowledgment and delivery confirmations	See Oman PASR



Appendix D QR code implementation

D.1 Method to generate Seller UUID

Generation of UUIDv5 must follow the steps as below:

1. Define the QR Version and Invoice Type as described above
2. Concatenate the values in a single String use the character "|", the resulting String value must follow the structure as below examples:
 - a. Examples for B2C Full and Simplified PINT
 - i. "1|1|SELLERNAME|VATTIN|DATE|TIMESTAMP|INVOICETOTALAMOUNT|VATTOTAL"
 - ii. "1.0|01|SellerA|OM1234567891|2026-05-13|18:31:00|302.340|14.520"
 - b. Examples for Profit Margin PINT invoice
 - i. "1|2|SELLERNAME|VATTIN|DATE|TIMESTAMP|TOTALAMOUNTDUDE"
 - ii. "1.0|02|SellerABC|OM1234567892|2026-05-13|18:30:00|1032.210"
3. Ensure all letters in the final string are UPPERCASE letters, this is important, UUIDv5 generation is CASE SENSITIVE
4. Generate the UUIDv5 using the FIXED namespace defined in Guideline and the resulted concatenated string

Code Examples:

UUIDv5 Generation - Python Expand source

```
import uuid

# 1. Define the required inputs
FIXED_NAMESPACE = uuid.UUID('e0bc4ac8-b025-46e5-a76d-0c893fc3027e')
qr_version = "1.0"
invoice_type = "01"
seller_name = "SELLERA"
vattin = "OM1234567891"
date = "2026-05-13"
timestamp = "18:31:00"
total_amount = "302.34"
vat_total = "14.52"

# 2. Construct the string, ensuring it is uppercase
input_string =
f"{qr_version}|{invoice_type}|{seller_name}|{vattin}|{date}|{timestamp}|{total_
amount}|{vat_total}".upper()

# 3. Generate the UUIDv5
generated_uuid = uuid.uuid5(FIXED_NAMESPACE, input_string)
```



UUIDv5 Generation - Javascript

```
import { v5 as uuidv5 } from 'uuid';

// 1. Define the required inputs
const FIXED_NAMESPACE = 'e0bc4ac8-b025-46e5-a76d-0c893fc3027e';
const qrVersion = "1.0";
const invoiceType = "01";
const sellerName = "SELLER-A";
const vattin = "OM1239222";
const date = "2026-05-13";
const timestamp = "12:00:00";
const totalAmount = "302.340";
const vatTotal = "14.520";

// 2. Construct the string, ensuring it is uppercase
const inputString =
`${qrVersion}|${invoiceType}|${sellerName}|${vattin}|${date}|${timestamp}|${totalAmount}|${vatTotal}`.toUpperCase();

// 3. Generate the UUIDv5
const generatedUuid = uuidv5(inputString, FIXED_NAMESPACE);
```

D.2 Method to generate the QR Code

Once the UUIDv5 is generated and all the values required for QR generation are available, the QR Code must be generated as described below:

To generate and print QR code encoded in Base64 format with up to 700 characters that must contain the fields specified in section 4 tables.

The QR code fields shall be encoded in Tag-Length-Value (TLV) format with the tag values specified in the “Tag” column of the adjacent table.

The TLV encoding shall be as follows:

- **Tag:** the tag value as mentioned above stored in one byte
- **Length:** the length of the byte array resulted from the UTF8 encoding of the field value. The length shall be stored in one byte.
- **Value:** the byte array resulting from the UTF8 encoding of the field value.

For every electronic tax invoice, the supplier must generate and print a QR code encoded in Base64.

- The Base64 string may not exceed 700 characters.
- The QR code content is defined in Tag–Length–Value (TLV)S format, as described below:



Item	Requirement
Tag	One byte containing the numeric tag value from the Table below.
Length	One byte indicating the length, in bytes, of the UTF-8-encoded Value
Value	UTF-8 text

Tag (1 Byte)	Length (1 Byte)	Value (L number of Bytes)
--------------	-----------------	---------------------------

What is a TLV (Tag - Length - Value) file format and how is it constructed?

QR code is the base64 encoded TLV. TLV is an encoding scheme used in many communication protocols to encode data. A TLV-encoded message has a defined structure which consists of 3 sections/parts. Those are:

Code of the message type (**T**) - 1 Byte

Message value length (**L**) - 1 Byte

Message value itself (**V**) - Variable

The Tag/Type and Length are of fixed sizes of 1 bytes while the value has a variable size.

As the general idea behind encoding is to transform abstract data into a stream of bits, using TLV, there are different sets of encoding rules that can be used according to the Abstract Syntax Notation Version 1 (ASN.1).

We are using a simple version of Basic Encoding Rules (BER).

TLV Field - Tag Mapping table

B2C Full and Simplified PINT

Field	Tag
QR Version	1
Invoice Type	2
Seller Name	3
VATIN of the Seller	4
Date	5
Time	6
Invoice total amount (with VAT)	7
VAT total	8



Seller UUID	9
-------------	---

Table 5 QR Code content TLV field definitions Simplified (and full) PINT

Profit Margin PINT Invoice

Field	Tag
QR Version	1
Invoice Type	2
Seller Name	3
VATIN of the Seller	4
Date	5
Time stamp	6
Total amount due	7
Seller UUID	8

Table 6 QR Code content TLV field definitions Profit Margin PINT Invoice

Tag-Length-Value (TLV) Creation**Example TLV Building – B2C Full and Simplified PINT:****Step 1: Create the hexadecimal representation:**

Field	Tag	Length	Example Value	Hexadecimal ("Value" is UTF-8 Encoded, So 1 = 31, where 31 is the UTF-8 of 1)
QR Version	1	3 (FIXED)	1.0 (FIXED)	0103312E30
Invoice Type	2	2 (FIXED)	01	02023031
Seller Name	3	7 (VARIABLE)	SellerA	030753656C6C657241
VATTIN	4	12 (FIXED)	OM1234567891	040C4F4D31323334353637383931
Date	5	10 (FIXED)	2026-05-13	050A323032362D30352D3133
Time Stamp	6	8 (FIXED)	18:31:00	060831383A33313A3030
Invoice Total Amount (with VAT)	7	7 (VARIABLE)	302.340	07073330322E333430
VAT Total	8	6 (VARIABLE)	14.520	080631342E353230
Seller UUID	9	36 (FIXED)	0627f995-487e-5384-9ceb-e484d8f9b2b3	092430363237663939352D343837652D353338342D396365622D653438346438663962326233

Final TLV Hex Representation:



```
0103312e3002023031030753656c6c657241040c4f4d31323334353637383931050a323032362d30352d31330608313
83a33313a303007073330322e333430080631342e353230092430363237663939352d343837652d353338342d39636
5622d653438346438663962326233
```

Step 2: Convert to Base64 Representation:

Resulted value: (Must not exceed 700 characters)

```
AQMxLjACAjAxAwDTZWxsZXJBBAxPTTEyMzQ1Njc4OTEFCjIwMjYtMDUtMTMGCD40jMxOjAwBwczMDluMzQwCAYxNC
41MjAJJDA2MjdmOTk1LTQ4N2UtNTM4NC05Y2VlWU00DRKOGY5YjIiMw==
```

Example TLV Building - Profit Margin PINT Invoice:**Step 1: Create the hexadecimal representation:**

Field	Tag	Length	Example Value	Hexadecimal ("Value" is UTF-8 Encoded, So 1 = 31, where 31 is the UTF-8 of 1)
QR Version	1	3 (FIXED)	1.0 (FIXED)	0103312E30
Invoice Type	2	2 (FIXED)	02	02023032
Seller Name	3	9 (VARIABLE)	SellerABC	030953656C6C6572414243
VATTIN	4	12 (FIXED)	OM1234567892	040C4F4D31323334353637383932
Date	5	10 (FIXED)	2026-05-13	050A323032362D30352D3133
Time stamp	6	8 (FIXED)	18:30:00	060831383A33303A3030
Total Amount Due	7	8 (VARIABLE)	1032.210	0708313033322E323130
Seller UUID	8	36 (FIXED)	8c978eb2-d3e8-5d88-bd12-a0132dd95b20	082438633937386562322D643365382D356438382D626431322D613031333264643935623230

Final TLV Hex Representation:

```
0103312e3002023032030953656c6c6572414243040c4f4d31323334353637383932050a323032362d30352d31330608
31383a33303a30300708313033322e323130082438633937386562322d643365382d356438382d626431322d6130313
33264643935623230
```

Step 2: Convert to Base64 Representation:

Resulted value: (Must not exceed 700 characters)

```
AQMxLjACAjAyAwlTWXsZXJBQkMEDE9NMTIzNDU2Nzg5MgUKMjYyNi0wNS0xMwYIMTg6MzA6MDAHCD40jMxOjAwBwczMDluMjYwCAYxNC
CCQ4Yzk3OGViMi1kM2U4LTvkODgtYmQxMi1hMDEzMmRkOTViMjA=
```

Considerations

- The Tag and Length are binary values of exactly one byte, therefore represented as Hex, example: "1" is "01" and "12" is "0C"
- The Value must also be converted to binary before adding to the byte array



- There should be no padding or separators between the TLV sets in binary array, the binary bytes should follow each other concurrently, the decoding is done by reading the length of each Tag from the second byte in each TLV pair.
- In order to encode Arabic Text into binary it is important to use UTF-8 Encoding

Code Example

The following code example illustrates how the base64 value can be retrieved, using the same values from above table describing the TLV for **Simplified and Full PINT**, note that the same logic can be followed in order to fulfil any other type of invoice or future types

```
Base64 Generation Example - Javascript
function getTLVForValue(tagNum, tagValue) {
    // T: TAG
    var tagBuf = Buffer.from([tagNum], 'utf8');

    // L: LENGTH
    var tagValueLenBuf = Buffer.from([tagValue.length], 'utf8')

    // V: VALUE
    var tagValueBuf = Buffer.from(tagValue, 'utf8');

    // TLV
    var bufsArray = [tagBuf, tagValueLenBuf, tagValueBuf]

    return Buffer.concat(bufsArray);
}

// 1. QR Version
var qrVersionBuf = getTLVForValue("1", "1.0");

// 2. Invoice Type
var invoiceTypeBuf = getTLVForValue("2", "01");

// 3. Seller Name
var sellerNameBuf = getTLVForValue("3", "SellerA");

// 4. VATTIN
var vattinBuf = getTLVForValue("4", "OM1234567891");

// 5. Date
var dateBuf = getTLVForValue("5", "2026-05-13");

// 5. Time Stamp
var timeStampBuf = getTLVForValue("6", "18:31:00");

// 7. Invoice Total Amount (with VAT)
var invoiceTotalBuf = getTLVForValue("7", "302.340");

// 8. VAT Total
var vatTotalBuf = getTLVForValue("8", "14.520");

// 9. Seller UUID
```



```
var sellerUuidBuf = getTLVForValue("9", "0627f995-487e-5384-9ceb-  
e484d8f9b2b3");  
  
// --- Combine all TLV buffers and encode to Base64 ---  
  
// (1) Create an array of all the generated buffers in order  
var tagsBufsArray = [  
    qrVersionBuf,  
    invoiceTypeBuf,  
    sellerNameBuf,  
    vattinBuf,  
    dateBuf,  
    timeStampBuf,  
    invoiceTotalBuf,  
    vatTotalBuf,  
    sellerUuidBuf  
];  
  
// (2) Concatenate all buffers into a single buffer  
var qrCodeBuf = Buffer.concat(tagsBufsArray);  
  
// (3) Convert the final buffer to a Base64 string  
var qrCodeB64 = qrCodeBuf.toString('base64');  
  
// Generate QR Code from the B64 string..
```